

Temporary Legal Protection as a Buffer against Immigration Enforcement: Employment, Income, and Poverty under DACA*

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Abstract

This paper examines the extent to which temporary legal protection buffers immigrants from the adverse effects of immigration enforcement, showing how overlapping immigration policies shape their economic outcomes. Using 2005-2019 American Community Survey data and a triple-differences framework, we compare likely undocumented Hispanic immigrants eligible for DACA to those narrowly ineligible. Enforcement worsens income and poverty among ineligible immigrants; DACA eligibility offsets roughly 75-90% of this penalty during the DACA period, with weaker impacts on labor supply. Falsification tests using naturalized citizens show no similar pattern while DACA was fully in force. These effects weaken after DACA's 2017 rescission attempt.

JEL Codes: J15, J61, K37

*An Online Appendix is available with this article. *Data Availability Statement:* The American Community Survey data are publicly available from IPUMS USA ([Ruggles et al., 2023](#)); enforcement policy data are compiled from the published sources cited in the text. Replication files will be posted upon publication. Contact: Samyam Shrestha (samyam@uga.edu). *Disclosure Statement:* Neither author received external funding for this research. Neither author has received financial support summing to \$10,000 or more in the past three years from any party with an interest in this work, holds a relevant position, or has relatives or partners with relevant interests. No other party had the right to review this paper prior to its circulation. This study uses publicly available, de-identified secondary data and did not require Institutional Review Board approval.

I. Introduction

Since the early 2000s, a procession of federal, state, and local enforcement policies has raised the real and perceived costs of living in the United States without authorization, with documented consequences for labor market behavior, poverty, and family life. These policies span several distinct types, from police-based measures such as arrests and deportations to employment-based measures such as work eligibility verification and service-based restrictions affecting access to driver’s licenses, education, and public benefits. At the same time, the federal government has periodically extended partial or temporary legal protection to specific subsets of the unauthorized immigrant population. The most politically and economically salient example of such legal protection is the Deferred Action for Childhood Arrivals (DACA), which since 2012 has provided temporary protection from deportation and work authorization to over 800,000 recipients who came to the United States as children.¹

While previous work has studied these two strands of immigration policy separately, this paper studies their interaction, asking whether DACA shielded its recipients from the consequences of living amid enhanced enforcement. We focus on labor market, income, and poverty outcomes both because they capture core dimensions of material well-being and because these are the margins along which each strand of policy has independently been shown to operate: enforcement depresses employment and earnings among likely unauthorized workers (East et al., 2023; Orrenius and Zavodny, 2015; Rubalcaba et al., 2024), while DACA eligibility has separately been shown to raise labor force participation and reduce poverty (Amuedo-Dorantes and Antman, 2016; Chin et al., 2026; Pope, 2016). To our knowledge, this is the first paper to test DACA’s effectiveness as a buffer against the downstream effects of enhanced enforcement.

To study this, we combine individual-level data from the American Community Survey (ACS) with a commuting-zone-level index of immigration enforcement intensity built from five enforcement instruments spanning 2005 to 2019. Following the DACA literature, we identify likely undocumented Hispanic immigrants and classify them as DACA-eligible or narrowly ineligible based on the program’s statutory age, arrival, and residency thresholds. We then estimate a triple-difference specification that interacts local enforcement intensity, DACA eligibility, and the DACA policy period, allowing us to isolate how the enforcement gradient facing DACA-eligible immigrants diverged from that facing the just-ineligible comparison group once the program took effect. Departing from previous literature, our empirical design emphasizes not just the change in economic outcomes for the DACA-eligible under DACA, but the interaction of their eligibility and the rollout of enhanced enforcement policy.

¹Other examples of such partial or temporary legal protection include Temporary Protected Status and the Deferred Enforced Departure.

We find that DACA eligibility meaningfully offset the adverse effects of enforcement on immigrants' economic outcomes throughout the years the program was fully in force. In commuting zones with higher enforcement intensity, DACA-eligible immigrants experienced smaller declines in income and a smaller rise in poverty relative to their just-ineligible counterparts, with a somewhat weaker but still positive protective effect on labor force participation. In our preferred estimates, DACA eligibility offset roughly 75-90% of enforcement's adverse effect on income and poverty during the years the program carried full legal force, a magnitude we quantify directly rather than infer from the sign of the underlying coefficients alone. These patterns emerge from a triple-differences design, show no evidence that enforcement affected eligible and ineligible immigrants differently before DACA took effect, and hold up across a wide range of alternative fixed-effects structures and sample and control-group constructions. They are also concentrated in the commuting zones where enforcement is most economically salient and where the largest share of the local population is plausibly exposed to both enforcement and DACA eligibility.

Two additional patterns reinforce this interpretation. The protective effect weakens substantially in years during which DACA's future was uncertain following the program's attempted rescission in 2017, suggesting that the credibility of legal protection, rather than formal eligibility alone, is what shields recipients from the indirect costs of enforcement. In addition, when we repeat the analysis on naturalized Hispanic citizens, a demographically similar group with no legal need for DACA and no comparable exposure to its protections, we find no equivalent buffering pattern in the pre-recession-attempt years. This indicates that our results are not simply capturing broader trends affecting Hispanic communities or local conditions correlated with enforcement. We find suggestive evidence that part of this protection operates through occupational sorting: because immigrant-intensive work is more exposed to enforcement, DACA-eligible individuals, whose legal status lets them move without added risk, shift away from such occupations as enforcement intensifies, while their ineligible counterparts remain in place. We also show that this protection is not driven by geographic relocation to lower-enforcement areas, indicating that DACA protects recipients in place rather than through selective residential sorting.

Our paper contributes to the growing body of research on the labor and poverty effects of immigration enforcement policies by identifying how these effects might be attenuated through protective policies like DACA. On the labor demand side, employment-based enforcement programs such as E-Verify have been shown to reduce the employment and earnings of likely unauthorized immigrants ([Amuedo-Dorantes and Bansak, 2012](#); [Orrenius and Zavodny, 2015](#)), and police-based programs generate similar effects: [Kostandini et al. \(2014\)](#) show that 287(g) agreements reduced immigrant presence in adopting counties, with farm wages, labor use, output, and profitability affected in ways consistent with resulting labor shortages. More recent work builds a broader consensus around negative labor market impacts for the unauthorized ([Amuedo-Dorantes and Antman,](#)

2022) that extend to US-born individuals as well (East et al., 2023; East and Velásquez, 2022). Not every labor supply response runs in the same direction, however: Rubalcaba et al. (2024) find that US-born Hispanic youth increase their labor force participation following enforcement shocks, likely stepping in as within-family substitutes for parents whose own labor supply enforcement has curtailed. These labor market costs carry over into material hardship: Amuedo-Dorantes et al. (2018) show that police-based enforcement directly increases poverty among families with US-born children and likely unauthorized parents, and related work documents similar consequences for safety net participation and household stability more broadly (Alsan and Yang, 2024).

We also build on the DACA literature by asking how DACA, a national policy, interacted with local variation in enforcement intensity, creating a protection wedge between DACA-eligible and just-ineligible immigrants that varies by geography and time. Among protective policies, DACA is the one most directly tied to labor market access, conferring work authorization alongside deportation relief. An existing literature shows that DACA increased labor force participation for those eligible, and reduced the likelihood of being in poverty (Amuedo-Dorantes and Antman, 2016, 2017; Pope, 2016), though a recent paper claims the effects are smaller or even zero (Tran, 2025), and contemporary work suggests DACA’s main effect on labor may have been compositional, with those eligible shifting into better and safer types of work (Chin et al., 2026). In this respect we join Amuedo-Dorantes and Arenas-Arroyo (2022), who study how sanctuary policies can alleviate the effects of enhanced interior enforcement, though our empirical design is more explicitly built to identify a policy interaction effect rather than the effect of a single protective policy in isolation.²

The remainder of the paper proceeds as follows. Section II provides background on the both immigration enforcement and DACA. Section III describes the data and Section IV outlines the empirical framework. Section V presents the results, then Section VI concludes.

II. Institutional Background

A. Immigration Enforcement in the Interior United States

Following the 9/11 attacks, policymakers at local, state, and federal levels sought ways to increase immigration controls, especially in the interior United States. A legal basis for some types of enhanced enforcement at the federal level had been previously codified in the Immigration Reform and Control Act of 1986 (IRCA) and the Illegal Immigration Reform and Immigrant Responsibility Act of 1996 (IIRAIRA). This meant that enforcement-minded policymakers required little more than commitments from local law enforcement agencies and funding for existing mecha-

²Amuedo-Dorantes and Arenas-Arroyo (2022) show that under enhanced enforcement, sanctuary policies can lead to higher self-petitions for domestic violence relief among the likely unauthorized.

nisms to enact certain enforcement measures (e.g. 287(g) agreements). Other efforts were piloted and expanded more recently, e.g. Secure Communities, a data sharing agreement launched in 2008 linking local arrests to federal immigration databases. Individual states also passed unique laws bundling multiple enforcement provisions, starting with Arizona’s SB 1070 in 2010. To capture the variation in these policies and their respective implementations, we follow previous literature, constructing an enforcement intensity index from a set of policies that varies over years and geography (Amuedo-Dorantes and Lopez, 2015; Watson, 2013). Here we briefly describe the policies included in the index: 287(g), Secure Communities, Omnibus Immigration Laws, and E-verify.

Some of the earliest-adopted enforcement efforts were 287(g) agreements, in which law enforcement agencies (LEAs) at the state or local level sought partnerships with the Department of Homeland Security (DHS) to perform federal immigration law functions.³ LEAs that applied chose between two models of 287(g) enforcement: the task-force model trained local officers to detain suspected unauthorized immigrants as part of routine police work, while the jail-based model allowed for screening the immigration status only of those booked into local jails. The task-force model represented a direct incursion of interior enforcement into communities, and its reliance on officer discretion generated claims of racial profiling (US Department of Justice, 2011). The first 287(g) agreements became active in 2002, and agreements were signed, denied, canceled, and rescinded throughout our study period.

Secure Communities (SC) was piloted in 2008, then rolled out nationwide by 2013. SC effectively automated the jail-based 287(g) model and mandated unilateral cooperation across the country. Under SC, Immigrations and Customs Enforcement (ICE) is notified whenever an arrestee’s fingerprints match to an unauthorized record within DHS, shifting prosecutorial discretion from local LEAs to a federal agency. SC resulted in the largest number of deportations of any policy in our index, about 454,000 between 2008 and 2014 (East et al., 2023). The program scope was narrowed and SC was formally renamed the Priority Enforcement Program (PEP) from 2015 to 2017, though removals continued under the same fingerprint-matching infrastructure.⁴ PEP-era removals added roughly 185,000 more deportations through FY2017, before the Trump administration reinstated the Secure Communities name in 2017 (TRAC Immigration, 2017).⁵

Beginning with Arizona’s SB 1070 in 2010, several states passed what are collectively known as Omnibus Immigration Laws (OILs). The typical omnibus law contained multiple policies aimed at reducing the presence of unauthorized immigrants in a given state, with many notably including

³287(g) refers to a section in the Immigration and Nationality Act. This law was passed in 1952, but section 287(g) was added later, as part of the IIRAIRA in 1996.

⁴The Transactional Records Access Clearinghouse (TRAC) records SC and PEP-related removals as a single continuous series.

⁵PEP maintained the biometric processing of SC, but DHS imposed guidance limiting ICE’s issuance of transfer and detainer requests to those convicted of “specifically enumerated crimes.” An archived comparison of SC and PEP is available at <https://www.ice.gov/pep>.

so called “papers, please” provisions, under which officers could demand proof of immigration status during lawful stops. These, along with 287(g) and SC are the primary police-based methods of enhanced enforcement in our index.

Distinct from enforcement through policing, E-verify targets unauthorized immigration through employment. Upon hiring a worker, firms in states requiring participation enter the worker’s information into the E-verify system. The worker’s information is then checked against records from the Social Security Administration (SSA) and DHS. If the system finds a discrepancy, it issues a tentative non-confirmation, and the worker may contest this finding before any adverse action is taken; only an unresolved final nonconfirmation creates legal risk for continued employment. While E-verify was available to employers in all states in 2003, participation was voluntary. However, starting with Arizona in 2007, several states passed laws requiring universal E-verify participation for all new hires. Research has shown that E-verify may have reduced state-wide populations of likely-unauthorized immigrants, or even agricultural workers generally, independent of immigration status. (Luo and Kostandini, 2022; Orrenius and Zavodny, 2016).

B. Deferred Action for Childhood Arrivals

President Obama announced the DACA program on June 15, 2012. The program targeted up to 1.7 million unauthorized immigrants who had entered the United States as children (Giuntella and Lonsky, 2020; Passel et al., 2014). Prior to 2010, Congress had attempted to create a legalization pathway for undocumented youth through several versions of the Development, Relief, and Education for Alien Minors (DREAM) Act, first introduced in 2001. However, the legislative momentum stalled by 2010 after Congress failed to reach agreement. Against this backdrop, DACA emerged as an executive action rather than a statutory reform. USCIS began accepting the first DACA applications in August 2012.

DACA provided two primary benefits. First, recipients received temporary and renewable protection from deportation (‘deferred action’), which confers lawful presence for certain purposes but does not grant lawful immigration status. Second, the program granted recipients an Employment Authorization Document (EAD). In addition to work authorization, an EAD enabled recipients to obtain a Social Security number, apply for state identification cards or driver’s licenses in many states, and access financial services such as credit cards, bank accounts, or loans that may have been previously unavailable to them.

DACA did not provide a path to citizenship, nor did it grant access to federal welfare programs or federal student aid. DACA recipients are eligible for the Earned Income Tax Credit, and several states extended state-funded Medicaid eligibility to DACA recipients.⁶ DACA status was initially

⁶These include California, Illinois, Massachusetts, Minnesota, New York, Oregon, and Washington.

granted for two years, after which recipients could apply for renewal. Renewal rates were high, with approximately 93 percent of recipients seeking renewal (Kuka et al., 2020), suggesting that many recipients expected the program to continue well beyond its initial two-year period.

To qualify for DACA, applicants had to meet several criteria: (1) lack lawful immigration status as of June 15, 2012; (2) have arrived in the United States before age 16; (3) be under age 31 as of June 15, 2012; (4) have continuously resided in the United States since June 15, 2007; (5) be currently in school, have graduated from high school, have obtained a GED certificate, or be an honorably discharged veteran of the Coast Guard or Armed Forces; and (6) have no felony convictions, significant misdemeanors, or three or more other misdemeanors. Applicants had to be at least 15 years old to apply at a cost of \$465, and approval rates were high, with more than 90 percent of processed applications approved (Pope, 2016).

In September 2017, the Trump administration announced its decision to rescind DACA. At the time, there were approximately 689,800 active DACA recipients, and an estimated 800,000 individuals had benefited from the program at some point (Bruno, 2021). Federal court injunctions beginning in early 2018 required USCIS to continue accepting renewal applications from existing recipients while the rescission was litigated, though initial applications from new applicants remained unavailable throughout this period. The rescission was challenged in federal court over the following years, and in June 2020 the Supreme Court ruled the rescission unlawful, though the full restoration of initial applications was not completed until subsequent litigation later that year. Our 2018 to 2019 uncertain period therefore corresponds to a period in which incumbent recipients, perhaps with some trepidation surrounding the program's future, could still renew their status, and new applicants faced continued uncertainty over whether DACA would remain available to them.

C. Interaction Between Immigration Enforcement and DACA

Interior immigration enforcement and DACA pull undocumented immigrants' economic outcomes in opposite directions. Enforcement raises the risk of detention and removal, lowers the expected return to working, and creates uncertainty that can discourage employment, mobility, and human capital investment. DACA counters this directly through temporary legal protection and work authorization, which lower deportation risk and open access to formal employment. Because DACA-eligible individuals remain fully exposed to enforcement while some share of the DACA-eligible population gains deferred action and work authorization by obtaining DACA status, the differential response of these two groups to changes in enforcement intensity offers a natural way to isolate how temporary legal protection interacts with enforcement, net of incomplete take-up among the eligible.

Three sources of variation make this comparison possible. Enforcement intensity varies across

locations and over time through the staggered adoption of enforcement policies. DACA eligibility varies across individuals based on predetermined characteristics. And DACA’s 2012 introduction, followed by rising policy uncertainty after its 2017 rescission attempt, splits the sample period into distinct regimes. Together, these three margins, geography and time in enforcement, individual eligibility, and policy period, let us test whether DACA-eligible individuals respond less to enforcement than their ineligible counterparts, and whether that gap narrows once DACA’s future is thrown into doubt. This differential response is the basis for our empirical strategy.

III. Data

A. Survey Data

Our primary data source is the American Community Survey (ACS), obtained from the Integrated Public Use Microdata Series ([Ruggles et al., 2023](#)). We use annual ACS samples from 2005 through 2019. The ACS provides detailed information on demographics, nativity, year of arrival, education, labor force participation, employment status, earnings, income, and program participation, which we use to construct proxies for DACA eligibility and measure our outcome variables.

The ACS does not identify the legal status of immigrants. Following [Passel and Cohn \(2014\)](#), we implement the residual method to approximate likely undocumented immigrants by removing individuals who are clearly documented based on observable characteristics. Specifically, we exclude citizens, Cuban-born individuals, veterans, and recent public assistance recipients,⁷ and retain the residual group of non-citizen, foreign-born immigrants who arrived in 1980 or later. In the main analysis, we focus on the Hispanic population because undocumented immigrants in the United States are disproportionately Hispanic, particularly from Mexico and Central America, and prior research shows that the majority of individuals potentially eligible for DACA come from these groups ([Passel et al., 2014](#)). Restricting the sample to Hispanic immigrants therefore improves the accuracy of the residual method in approximating the undocumented population and aligns with the demographic composition of the population most likely affected by DACA. This approach reduces, but does not eliminate, misclassification of legal status, as some residual individuals remain documented through channels the method cannot observe, while some undocumented immigrants are inadvertently excluded. We also impose the program’s education requirement by retaining individuals who have either completed at least a high school degree or are currently enrolled in school.

⁷We define ‘recent public assistance recipients’ as individuals reporting nonzero income from Supplemental Security Income or other public/welfare cash assistance in the prior year. This is distinct from SNAP/food stamp receipt, which we use only as an outcome variable, so this exclusion criterion does not mechanically overlap with our SNAP outcome.

Using the restricted sample, we construct a proxy for DACA eligibility based on statutory criteria. Individuals are classified as potentially eligible if they (i) arrived in the United States before age 16, (ii) were younger than 31 as of June 2012, and (iii) had resided in the United States for at least five years by 2012. Age at arrival is inferred from current age and year of immigration, and age eligibility is calculated using birth year and quarter of birth. Because the ACS does not contain information on criminal history, which is itself a statutory requirement for DACA eligibility, or on continuous residence since June 2007, this measure captures potential rather than confirmed eligibility.

We then construct the analysis sample to include both DACA-eligible individuals and a near-eligible comparison group. Eligible individuals satisfy all three criteria above. The comparison group satisfies the age-as-of-2012 criterion (which is shared by both groups by construction given our sample restrictions) but fails either the arrival-age or residence-duration requirement.⁸ To improve comparability, we further restrict the sample to individuals ages 18 to 38, those whose implied age in 2012 is less than 31, and those who arrived in the United States at age 17 or younger, so that neither discontinuity variable is left unbounded. These restrictions are chosen to align closely with the statutory DACA eligibility thresholds, ensuring that both groups are of working age, would have been under 31 as of June 2012, and arrived near the age-16 cutoff, thereby keeping the comparison group as similar as possible to the eligible group while remaining just outside the program's criteria.

We assign individuals to local labor markets at the commuting zone (CZ) level. Because ACS respondents are identified at the PUMA level, we use the crosswalk from [Autor and Dorn \(2013\)](#), harmonized to the 1990 CZ definitions ([Tolbert and Sizer, 1996](#)). We apply the 2000 PUMA crosswalk for 2005 to 2011 and the 2010 PUMA crosswalk for 2012 onward.⁹ Individuals are fractionally allocated across CZs according to PUMA population shares, and all estimates use ACS person weights adjusted by the PUMA to CZ share.¹⁰

⁸Because the residence requirement is tied to June 2007, comparison-group members who fail only this requirement (immigrants who arrived as children after 2007) enter our 18 to 38 age window gradually, as they turn 18, and are therefore concentrated in the post-2012 period. The comparison group thus shifts toward more recent arrivals over time, which could confound our estimates if recent arrivals respond differently to enforcement. To address this, Section C re-estimates all specifications restricting the comparison group to individuals who fail only the arrival-age requirement: those who arrived at ages 16 or 17 but satisfy the residence requirement, and whose composition is therefore stable across periods.

⁹The Census Bureau revised PUMA boundaries in 2012, necessitating separate crosswalks for the two periods.

¹⁰This procedure may increase the number of observations because individuals can map to multiple CZs, but the effective sample size is unchanged. In our regression results, we report both the total number of observations and the number of unique individual-years.

B. Data on Immigration Enforcement

We compile data on immigration enforcement timing from several sources. Specifically, we collect information on the timing of 287(g) program adoption at the county and state levels from [Kostandini et al. \(2014\)](#), E-Verify implementation from [Orrenius and Zavodny \(2015\)](#), state-level omnibus immigration laws from [Allen and McNeely \(2017\)](#) and [Luo and Kostandini \(2023\)](#), and sanctuary policy adoption from [Martínez-Schuldt and Martínez \(2019\)](#), where for each county we identify the earliest adoption date of any sanctuary policy at the city, county, or state level. We verify these data against official records from ICE and the DHS.

We follow [Amuedo-Dorantes et al. \(2018\)](#) to create our immigration enforcement intensity variable. We first construct an intermediate variable with the following formula:

$$(1) \quad ENF_{z,t}^k = \frac{1}{N_{z,0}} \sum_{c \in z} \frac{1}{12} \sum_{j=1}^{12} \mathbb{1}(E_{c,j,t}^k) P_{c,0}$$

where $\mathbb{1}(E_{c,j,t}^k)$ is an indicator function that is equal to 1 if immigration enforcement policy k is active in county c in CZ z during month j in year t .¹¹ $P_{c,0}$ and $N_{z,0}$ are measured using 2000 population, the baseline year for our index. The indicator $\mathbb{1}(E_{c,j,t}^k)$ reverts to zero in any county-month following a policy's cancellation or rescission; unlike Secure Communities/PEP, which we treat as a continuous enforcement-infrastructure series, 287(g) agreements and other components are coded as active only while formally in effect. $ENF_{z,t}^k \in [0, 1]$ thus denotes the baseline-population-weighted fraction of CZ z where policy k was active in year t . Our analysis includes all 710 CZs across the contiguous United States.

We create this measure for five policies: the Immigration and Nationality Act Section 287(g) at the state and county levels, Secure Communities, E-Verify, and Omnibus Immigration Bills. We treat sanctuary policy exposure as a control variable rather than a component of the enforcement index, as sanctuary policies work in the opposite direction by limiting local cooperation with federal immigration enforcement.

We then aggregate across policies as:

$$(2) \quad ENF_{z,t} = \sum_k ENF_{z,t}^k$$

to create our aggregated immigration enforcement index. For ease of interpretation, we use the standardized version of this index, normalized to have mean zero and standard deviation one across commuting zones and years in our sample.

¹¹For Secure Communities specifically, we consider a month as treated if a county implemented the program on or before the 15th of the month.

Figure 1 illustrates the geographic distribution and temporal evolution of the enforcement index across commuting zones for selected years. In 2005, enforcement intensity was low and concentrated in a small number of CZs, primarily in Florida, Alabama, and the Southwest. By 2010, enforcement had spread considerably, with notably higher intensity in Arizona, new areas of the Southeast, and scattered CZs in the interior. The 2015 map reflects a near-peak of interior enforcement activity, driven by the cumulative expansion of the enforcement instruments in our index through the early-to-mid 2010s. 2019’s map shows some minor variation from 2015, mainly driven by a renewed interest in 287(g) agreements during the first Trump administration. Together, these maps highlight the substantial spatial and temporal variation in enforcement intensity that underlies our empirical strategy.

C. Descriptive Statistics

Table 1 presents summary statistics by DACA eligibility and period. The eligible and non-eligible groups differ in characteristics that reflect the DACA eligibility criteria. Eligible individuals arrive at much younger ages (around 7-8 years old on average, compared to 15-16 for ineligible individuals) and have spent more years in the United States. They are slightly younger overall in the pre-period, though ages converge in the post-period. In terms of family structure, eligible individuals are somewhat more likely to be single and have fewer children. Educational attainment is broadly similar, with most individuals having completed high school, and only small differences across groups. These patterns are consistent with the institutional design of DACA and highlight systematic but expected compositional differences between the two groups.

Turning to economic outcomes, DACA-eligible and ineligible individuals look broadly similar across key labor market and economic outcomes, though eligible individuals start from slightly lower baseline levels. In the pre-DACA period, eligible individuals have somewhat lower labor force participation (70% vs. 75%) and employment rates (61% vs. 68%), as well as lower wage and total income. However, conditional employment rates are very similar across groups, and poverty measures are nearly identical, with comparable shares below the poverty line and at higher poverty thresholds.

In the post-DACA period, both groups experience improvements in labor market outcomes and income, but the gains are more pronounced for DACA-eligible individuals. Labor force participation rises to 78% for eligible individuals compared to 72% for ineligible individuals, and employment shows a similar pattern. Income increases for both groups, with eligible individuals slightly surpassing ineligible individuals in average earnings. Poverty rates decline for both groups, with somewhat larger reductions among the eligible, while SNAP participation increases modestly for both. Overall, the two groups remain similar across most dimensions, but the descriptive patterns

point to modest relative improvements for DACA-eligible individuals after 2012.

Figures A1-A3 plot the corresponding annual means for employment, income, and poverty outcomes by eligibility status. Eligible individuals' labor force participation and income rise relative to their ineligible counterparts after 2012, although these unadjusted series do not account for local enforcement exposure or the controls in our empirical strategy.

IV. Empirical Strategy

A. Empirical Specification

To examine whether DACA eligibility attenuated the effects of immigration enforcement, our main estimating equation is

$$\begin{aligned}
 Y_{izt} = & \alpha + \beta_1 ENF_{zt} + \beta_2 DACA_i + \beta_3 (ENF_{zt} \times DACA_i) \\
 & + \beta_4 (ENF_{zt} \times Active_t) + \beta_5 (ENF_{zt} \times Uncertain_t) \\
 (3) \quad & + \beta_6 (DACA_i \times Active_t) + \beta_7 (DACA_i \times Uncertain_t) \\
 & + \beta_8 (ENF_{zt} \times DACA_i \times Active_t) + \beta_9 (ENF_{zt} \times DACA_i \times Uncertain_t) \\
 & + X'_{izt} \gamma + W'_{zt} \omega + \nu_z + \lambda_t + \eta_{tc} + \varepsilon_{izt}
 \end{aligned}$$

where Y_{izt} denotes an outcome for individual i residing in CZ z in year t . The variable ENF_{zt} measures immigration enforcement intensity in CZ z and year t . The indicator $DACA_i$ equals one if individual i meets the DACA eligibility criteria and zero otherwise, while $Active_t$ equals 1 for the period 2013-17 where DACA was fully enacted. $Uncertain_t$ equals 1 for years 2018-19 where the status of DACA was uncertain.¹²

The vector of individual-level controls, X_{izt} , includes age (and its higher-order polynomials), male dummy, education level indicators, marital status indicators, number of children in the household, and English proficiency dummies. The vector W_{zt} includes CZ-level controls, including a Bartik-style immigrant-specific shift-share control for local labor demand shocks (Bartik, 1991; East et al., 2023; Goldsmith-Pinkham et al., 2020),¹³ minimum wage, sanctuary policy status,¹⁴

¹²DACA was announced in June 2012 and the program began accepting applications in August 2012. Most initial approvals occurred toward the end of 2012 and into early 2013. In the baseline specification, we treat 2012 as a pre-treatment period to avoid partial exposure within the year.

¹³The Bartik index is constructed from American Community Survey microdata, 2005-2019, over the full working-age sample. Employment is allocated from PUMAs to commuting zones using person weights adjusted by the PUMA-to-CZ crosswalk allocation factor. Baseline industry shares $\omega_{c,s}$ are computed from pooled 2005-2006 CZ-industry employment, and leave-one-out national industry growth $g_{-c,s,t}$ is computed by excluding each CZ's own employment from the national total, relative to the same 2005-2006 baseline: $B_{c,t} = \sum_s \omega_{c,s} \cdot g_{-c,s,t}$.

¹⁴We construct a CZ-level measure of sanctuary policy exposure as the share of the population residing in treated counties, weighted by county population in 2000. For each county, we identify the earliest adoption date of any sanc-

and quadratic trends in the 2000-2006 change in state-level housing price index to capture local economic conditions and cost-of-living dynamics that may independently affect immigrant outcomes.¹⁵

We include CZ fixed effects (ν_z) to absorb time-invariant local characteristics, year fixed effects (λ_t) to capture common macroeconomic shocks, and year-by-arrival-cohort fixed effects (η_{tc}) to flexibly control for differences across immigrant arrival cohorts. Regressions are weighted using the product of ACS person weights and PUMA-to-CZ crosswalk allocation shares to maintain population representativeness given the fractional assignment of individuals across commuting zones, and standard errors are clustered at the CZ level to account for serial correlation in enforcement exposure.

The coefficient β_1 captures the relationship between enforcement intensity and outcomes for just-ineligible immigrants in the pre-DACA period, β_2 captures the baseline difference between DACA-eligible and nearly ineligible individuals, β_3 measures how the relationship between enforcement intensity and outcomes differs by DACA eligibility status in the pre-DACA period, β_4 captures effect of enforcement intensity changes during the period 2013-2017 when DACA was stable, and β_5 captures the same effect during DACA’s uncertain period 2018-2019. β_6 captures the change in DACA eligibility’s baseline effect during the period 2013-2017 when DACA was stable, and β_7 captures the same change during DACA’s uncertain period, 2018-2019. Finally, β_8 and β_9 , the coefficients of interest for this paper, describe the differential (mitigating) effect of DACA eligibility on the impact of enforcement intensity, over the stable and uncertain periods, respectively.

The coefficients should be interpreted as intention-to-treat (ITT) effects rather than average treatment-on-the-treated (ATT) effects. The variable $DACA_i$ captures eligibility for the program rather than actual participation. Because not all eligible individuals applied for or received DACA status, the estimated coefficients reflect the effect of DACA eligibility in areas with different levels of immigration enforcement. Consequently, the ITT estimates represent attenuated versions of the treatment effects among actual recipients. We consider additional strategies to uncover estimates “closer” to the true ATT in section V.

tuary policy at the county, city, or state level, and determine annual treatment based on whether the policy is in effect for at least six months of a given year. We then aggregate to the CZ level by summing total and treated populations across counties within each CZ-year, and define exposure as the ratio of treated population to total population.

¹⁵We construct this variable as follows: we construct the population-weighted average state-level FHFA house price index within each state, computed for 2000 and 2006, where the percentage change defines a time-invariant state-level housing boom exposure. This shock is interacted with linear and quadratic time trends to flexibly capture differential housing cycle trajectories across states.

B. Identification and Validity

Identification relies on a difference-in-difference-in-differences (DDD) type framework that compares changes in outcomes between DACA-eligible and nearly ineligible Hispanic immigrants across commuting zones differentially exposed to immigration enforcement, before and after the implementation of DACA. DDD designs allow for flexible group-specific and eligibility-specific trends by differencing out time-invariant and group-invariant sources of bias (Gruber, 1994; Olden and Møen, 2022; Ortiz-Villavicencio and Sant’Anna, 2025). With continuous enforcement intensity, the DDD design tests whether the relationship between ENF_{zt} and outcomes changes after 2012 differentially for DACA-eligible immigrants. The triple-interaction coefficients β_8 and β_9 in Equation (3) capture this mitigating effect.

The key identifying assumption is that, in the absence of DACA, the enforcement gradient of the DACA-eligible minus nearly ineligible outcome gap is stable across pre-2012 years relative to the last pre-DACA year, t_0 . Formally, letting Y_{izt}^0 denote potential outcomes in the absence of DACA:

$$(4) \quad \begin{aligned} & \frac{\partial}{\partial ENF} E[Y_{izt}^0 \mid DACA_i = 1, ENF_{zt}] - \frac{\partial}{\partial ENF} E[Y_{izt}^0 \mid DACA_i = 0, ENF_{zt}] \\ &= \frac{\partial}{\partial ENF} E[Y_{iz,t_0}^0 \mid DACA_i = 1, ENF_{z,t_0}] - \frac{\partial}{\partial ENF} E[Y_{iz,t_0}^0 \mid DACA_i = 0, ENF_{z,t_0}] \end{aligned}$$

for all pre-2012 years $t \neq t_0$, where t_0 denotes the last pre-DACA year. The condition states that the enforcement gradient of the DACA-eligible minus just-ineligible outcome gap in any pre-treatment year t equals that same gradient in the base year t_0 , meaning this enforcement slope does not vary systematically across pre-policy years. This restriction is the continuous-treatment analogue of the DDD parallel trends condition emphasized in Olden and Møen (2022) and the DDD conditional parallel trends assumption in Ortiz-Villavicencio and Sant’Anna (2025).

Intuitively, absent DACA, the difference in pre-trends between DACA-eligible and just-ineligible immigrants would have been the same in high- and low-enforcement areas, up to a constant difference k . Under this parallel-trends-in-triple-differences assumption, the coefficient β_8 in Equation (3) can be interpreted as the causal effect of DACA in mitigating the effects of immigration enforcement.

Importantly, consistent with Olden and Møen (2022) and Ortiz-Villavicencio and Sant’Anna (2025) this condition *does not* require parallel trends in the levels of outcomes with respect to enforcement intensity. Identification would be violated if the differential effect of enforcement between DACA-eligible and just-ineligible immigrants changed over time prior to 2012. Accordingly, the identifying variation comes from changes in the enforcement gradient of the eligible-ineligible outcome gap after DACA, rather than from parallel trends in enforcement exposure it-

self.

We assess the plausibility of this identifying assumption using an event-study specification that replaces $Active_t$ and $Uncertain_t$ with a set of year indicators interacted with ENF_{zt} and $DACA_i$. Specifically, we estimate:

$$\begin{aligned}
(5) \quad Y_{izt} = & \alpha + \beta_1 ENF_{zt} + \beta_2 DACA_i + \beta_3 (ENF_{zt} \times DACA_i) \\
& + \sum_{\ell \neq -1} \theta_\ell (ENF_{zt} \times DACA_i \times \mathbf{1}\{t - t_0 = \ell\}) \\
& + \sum_{\ell \neq -1} \kappa_\ell (ENF_{zt} \times \mathbf{1}\{t - t_0 = \ell\}) + \sum_{\ell \neq -1} \rho_\ell (DACA_i \times \mathbf{1}\{t - t_0 = \ell\}) \\
& + X'_{izt} \gamma + \nu_z + \lambda_t + \eta_{tc} + \varepsilon_{izt}.
\end{aligned}$$

where t_0 denotes 2013, the first year DACA was implemented, so that $\ell = -1$ corresponds to 2012, the last pre-DACA year and the omitted reference category. The identifying assumption implies that the coefficients on all pre-DACA leads (θ_ℓ for $\ell < 0$) should be statistically indistinguishable from zero, indicating no pre-DACA changes in the enforcement gradient of the DACA-eligible minus just-ineligible outcome gap.

V. Results

This section presents our main empirical findings. We begin by documenting how DACA eligibility shaped the relationship between enforcement intensity and outcomes across three margins: labor force participation and employment, income, and poverty. We then quantify the size of this buffering effect as the share of enforcement's adverse effect on ineligible immigrants that is offset by DACA eligibility. Next, we discuss how imperfect measurement of legal status and incomplete DACA take-up bias our estimates toward zero and report results from subsamples with higher take-up rates as a partial correction. We then subject these findings to a series of checks: alternative specifications, alternative sample and control group constructions, and a falsification test using naturalized citizens. We close by examining the specific channels through which DACA's buffering effect operates.

A. Main Results

1. Employment

We begin by examining employment outcomes, as these represent the most immediate margin through which immigration enforcement and DACA are likely to operate. Immigration enforce-

ment increases the risk of detention and removal, which can affect individuals' willingness and ability to participate in the labor market, particularly in formal employment. In contrast, DACA provides work authorization and reduces deportation risk, directly altering access to legal employment opportunities. Labor force participation and employment are thus natural first-order outcomes, preceding downstream impacts on income and poverty.

Table 2 presents the results based on Equation (3). The outcome variables include three indicators: (i) a binary variable equal to one if the individual participates in the labor force, (ii) a binary variable equal to one if the individual is employed in the full sample, and (iii) a binary variable equal to one if the individual is employed conditional on labor force participation. All specifications include the full set of fixed effects, individual and CZ-level controls, and weighting and clustering choices described in Section IV.

Across all three specifications in Table 2, the coefficient on enforcement is small in magnitude and statistically insignificant, with point estimates ranging from -0.008 to 0.002, implying at most a 0.8 percentage point change in outcomes for a one standard deviation increase in the enforcement index for pre-DACA years.¹⁶ The interaction between enforcement and DACA eligibility is also close to zero across columns (-0.002 to -0.008), suggesting economically negligible differences in how enforcement relates to employment outcomes for eligible versus ineligible individuals in the pre-policy years. Similarly, the interaction between enforcement and the post-2017 period is small (0.002 to 0.007) and imprecisely estimated, indicating limited changes in the enforcement gradient during that period.

In contrast, the interaction between DACA eligibility and the DACA period is economically meaningful and precisely estimated. The coefficients range from 0.025 to 0.048, corresponding to increases of 2.5 to 4.8 percentage points in labor force participation and employment among eligible individuals relative to their ineligible counterparts during the DACA period. These magnitudes are comparable to prior estimates of DACA's labor market effects ([Amuedo-Dorantes and Antman, 2017](#); [Pope, 2016](#)).

The primary coefficient of interest, the triple interaction term, is positive across all outcomes, with magnitudes between 0.008 and 0.016. The estimate for labor force participation is statistically significant at the 10 percent level and implies that a one standard deviation increase in enforcement is associated with a 1.4 percentage point higher likelihood of labor force participation among DACA-eligible individuals during the DACA period. The corresponding estimates for employment

¹⁶One difference from related work is the construction of the outcome variable. [East et al. \(2023\)](#) measure employment of each group as a share of the local area's total population, an aggregate measure that can decline either because individuals stop working or because they leave the area, although they find little evidence that Secure Communities changed local population. Our labor force participation and employment measures are instead defined among individuals observed in the ACS in each year, so they capture changes in employment among those who remain rather than changes in the size of the local workforce.

outcomes are smaller and imprecisely estimated. In the post-2017 period, the triple interaction terms are close to zero (0.003 to 0.006) and not statistically distinguishable from zero.

This pattern suggests that DACA's protective effect operates mainly on the extensive margin of labor force participation rather than on employment conditional on participation. The labor force participation estimate is statistically significant only at the 10 percent level and this interpretation should be treated as suggestive rather than established. As Appendix Table A1 shows, the total enforcement gradient facing the just-ineligible group during the DACA period, combining the main enforcement coefficient with the relevant DACA-period interaction, remains small and statistically insignificant for employment outcomes. This indicates that enforcement's costs to this group are not primarily paid through reduced employment probability among those who remain in our sample. As we show in the next two sections, these costs instead surface in income and poverty outcomes, consistent with enforcement depressing the quality and formality of work, and the returns to working, more than it depresses the likelihood of working at all.

Event-study estimates show no evidence of diverging pre-trends for employment or employment conditional on labor force participation (Appendix Figure 2).¹⁷ The pre-trend test is rejected for labor force participation ($p = 0.002$), however, so we interpret that estimate with caution.

2. Income

Table 3 reports the effects of immigration enforcement and DACA eligibility on income outcomes, conditional on being employed. Focusing on employed individuals isolates earnings responses from changes in employment.¹⁸ We examine two measures, wage and salary income and total personal income from all sources, each in levels and in logs: levels capture dollar changes, while logs capture proportional changes and reduce the influence of very high earners.

The interaction between enforcement and the DACA period is negative and statistically significant across all specifications, with point estimates of about -710 dollars for wage income and -499 dollars for total income, alongside declines of 2.5 to 4.1 percent in log income measures. These results indicate that, among ineligible individuals, higher enforcement intensity during the DACA period is associated with economically meaningful reductions in earnings.

In contrast, the triple interaction term for the DACA period is positive and statistically significant across all outcomes, with magnitudes of roughly 667 to 688 dollars in levels and 2.9 to 4.1 percent in logs. This implies that DACA-eligible individuals experience higher income relative to ineligible individuals in areas with greater enforcement exposure during this period. In the post-2017 period, the corresponding triple interaction terms are small in magnitude and not statistically

¹⁷Joint F-tests of the pre-treatment event-time interactions fail to reject the null of parallel trends prior to DACA for employment ($p = 0.471$) and employment conditional on labor force participation ($p = 0.214$).

¹⁸Appendix Table A2 shows that the triple-interaction estimates remain positive when the employment condition is removed, though only the log wage income estimate remains statistically significant.

distinguishable from zero across all specifications, indicating no clear differential relationship between enforcement and income for DACA-eligible individuals in this later period. To test this change directly, we estimate the difference between the two triple-interaction coefficients, $\beta_8 - \beta_9$ (Appendix Table A3). The difference is positive and statistically significant for wage income, log wage income, and total income, confirming that the buffering effect on earnings weakened after 2017; for log total income, it is positive but imprecisely estimated.

The magnitude of this offset is meaningful relative to the underlying enforcement penalty. Against average pre-DACA wage income of roughly 15,200 dollars in our sample (Table 1), the total enforcement slope for the just-ineligible group during the DACA period is $-\$702$ (Table A4, Panel A), a loss of roughly 4.6 percent of baseline earnings for a one standard deviation increase in enforcement. The corresponding eligible-ineligible difference in this active-period slope, combining the triple interaction with the underlying double interactions ($\beta_3 + \beta_8$), is approximately $\$531$ (Table A4, Panel C minus Panel A). This suggests that the differential associated with DACA eligibility offsets a substantial share, though not all, of the negative enforcement gradient observed among the just-ineligible group during the years the program was fully in force.

This intensive-margin offset is consistent with prior evidence that enforcement's earnings penalty to unauthorized workers operates through the terms of employment rather than through job loss alone, as Amuedo-Dorantes and Bansak (2012) and East et al. (2023) find for E-Verify and Secure Communities respectively, and it points to the same occupational channel documented by Chin et al. (2026), whose finding that DACA eligibility shifts recipients into more formal, better-paying jobs offers a direct explanation for why legal status would raise earnings conditional on employment rather than the likelihood of holding a job at all. The fading of this offset after 2017 is consistent with evidence that DACA's labor market gains eroded under the policy uncertainty that followed the rescission announcement (Amuedo-Dorantes and Wang, 2026).

The dynamic pattern of these effects, shown in Figure 3, indicates that the divergence between eligible and ineligible immigrants' income opens up only after 2012 and contracts back toward zero following the 2017 rescission attempt, consistent with the buffering effect strengthening while DACA is fully in force and weakening once its legal status becomes uncertain.¹⁹

¹⁹Pre-treatment joint F-tests do not reject parallel trends for any income outcome: wage income ($p = 0.489$), log wage income ($p = 0.115$), total income ($p = 0.447$), and log total income ($p = 0.117$).

3. Poverty

Table 4 focuses on differences in poverty and welfare outcomes.²⁰ The interaction between DACA eligibility and the DACA period is generally negative across poverty measures, with statistically significant reductions of about -1.9 percentage points for both the 125 percent and 150 percent poverty thresholds. These estimates indicate that, during the DACA period, eligible individuals experienced lower poverty exposure relative to their ineligible counterparts. In contrast, the corresponding interaction in the post-2017 period shows a statistically significant increase of 2.8 percentage points in the probability of being below the poverty line, while other estimates are smaller and imprecisely estimated.

The triple interaction terms show that these differences vary with enforcement exposure. During the DACA period, the estimates are negative and statistically significant for several outcomes, including reductions of about -1.9 percentage points in the likelihood of being below the poverty line and -2.0 percentage points for the 125 percent threshold, along with a -1.7 percentage point decline in SNAP participation. The estimate for the poverty ratio is positive (3.25) and marginally significant, indicating improved relative income. In the post-2017 period, the triple interaction terms are generally smaller, though statistically significant reductions persist for the 125 percent poverty threshold (-2.2 percentage points) and SNAP participation (-1.9 percentage points), while other estimates are not statistically distinguishable from zero. Directly comparing the two periods, the difference between the triple-interaction coefficients, $\beta_8 - \beta_9$, is negative and statistically significant for being below the poverty line, indicating that the buffering effect on poverty weakened after 2017, but is small and imprecisely estimated for the remaining poverty and SNAP outcomes (Appendix Table A3).

These poverty results provide downstream evidence that the income gains documented in Section 2 are accompanied by material improvements in household wellbeing. Against a pre-DACA poverty rate of roughly 23 to 24 percent below the poverty line in our sample (Table 1), a 1.9 percentage point reduction associated with a one standard deviation increase in enforcement represents a reduction of about 8 percent relative to baseline, a nontrivial improvement given that this estimate reflects only the differential response to enforcement intensity rather than DACA's full effect. Our results align with Amuedo-Dorantes et al. (2018)'s finding that interior enforcement raises poverty among families with U.S.-born children and likely unauthorized parents, and with Amuedo-Dorantes and Antman (2016)'s finding that DACA eligibility on its own lowers poverty

²⁰SNAP receipt is measured at the household level in the ACS and is assigned to every household member regardless of that individual's own program eligibility. Neither DACA recipients nor other undocumented immigrants are personally eligible for federal SNAP benefits, so any household-level receipt in our sample is necessarily driven by other household members, such as a US-citizen child or a citizen or lawfully present spouse. We do not restrict our sample to mixed-status households, so this measure captures household-level food assistance receipt rather than the sampled individual's own eligibility.

exposure.

The relative decline in SNAP participation is consistent with reduced material need among DACA-eligible households, though we interpret it cautiously, since neither group's total enforcement gradient in SNAP participation is individually significant during the DACA period (Appendix Table A5). Because fear-driven reductions in take-up among the just-ineligible group (Alsan and Yang, 2024; Watson, 2014) would, if anything, bias this estimate upward, they are unlikely to explain it. As with income, the event-study estimates show no systematic pre-trends in poverty or SNAP outcomes (Figure 4).²¹

Taken together with Sections 1 and 2, these results indicate that enforcement's costs to the just-ineligible group appear primarily in earnings and material hardship rather than in whether they work at all, and that DACA eligibility offset much of these costs while the program was fully in force.

4. How Large is the Buffering Effect?

The results in the preceding sections establish that enforcement worsens economic outcomes for DACA-ineligible individuals while DACA eligibility substantially offsets these effects, but they do not by themselves indicate how large this offset is. To quantify this, for each outcome and the DACA-active period we compute the share of the total enforcement effect on the ineligible group that is absent for the eligible group.²² Appendix Table A6 reports these implied offsets for all outcomes.

The most precisely estimated offsets are for below-poverty-line status and log wage income, at 91.3% (95% CI: 65.1-117.6%) and 90.3% (95% CI: 46.6-133.9%), respectively. The remaining income and poverty measures, wage income, total income, log total income, poverty ratio, and below-125% poverty, yield point estimates in the same 75-92% range, though with wider confidence intervals. Labor force participation and overall employment show directionally similar point estimates, 98.2% and 82.3%, but their confidence intervals are wide enough that these should be read as suggestive rather than precise. Employment conditional on labor force participation, below-150% poverty, and SNAP participation yield confidence intervals that include zero or are otherwise uninformative, and we do not interpret these as evidence of any offset. We similarly do not report offset estimates for the post-2017 uncertain period, since the underlying enforcement effect on the

²¹Pre-treatment joint F-tests fail to reject parallel trends for the poverty-to-income ratio ($p = 0.802$), income below the poverty line ($p = 0.423$), income below 125% of the poverty line ($p = 0.599$), income below 150% of the poverty line ($p = 0.412$), and SNAP participation ($p = 0.111$).

²²Specifically, we compute $1 - \frac{\beta_1 + \beta_3 + \beta_4 + \beta_8}{\beta_1 + \beta_4}$, where the numerator is the total enforcement effect for DACA-eligible individuals during the active DACA period and the denominator is the corresponding total effect for ineligible individuals, using coefficients from the specification in Equation (3). Confidence intervals are computed via the delta method to account for the fact that this is a nonlinear function of several jointly estimated coefficients.

ineligible group becomes imprecisely estimated for every outcome during that period, making the implied ratio uninformative regardless of DACA eligibility.

We do not interpret the gap between these offsets and full protection as evidence of a residual enforcement penalty for DACA-eligible individuals. For most outcomes, the confidence intervals include 100 percent, and the total enforcement effect on the eligible group during the DACA period is small and statistically insignificant (Appendix Tables A4 and A5, Panel C). Moreover, our estimates are intention-to-treat effects: because many eligible individuals never obtained DACA and remained exposed to enforcement, the offset among actual recipients is likely larger, consistent with the larger triple-interaction estimates we find in high-takeup subsamples (Section B). To the extent protection was incomplete, it may reflect spillovers from enforcement against unprotected household members, local labor market effects that reach all workers in immigrant-intensive sectors, or the temporary nature of DACA status itself.

Overall, these offsets indicate that during the years DACA carried full legal force, eligibility eliminated the large majority, roughly 75 to 90% by our most reliable estimates, of enforcement's adverse effect on income and poverty. This near-complete protection is what we mean when we describe DACA as a buffer against enforcement.

B. Intention-to-Treat Effects and Heterogeneity

Next, we consider two sources of potential bias toward zero in our DDD estimates. The first relates to the imperfect measurement of immigration status in the ACS. Using the observable DACA eligibility criteria, with no restrictions on ethnicity or national origin, the noncitizen sample is estimated to contain 40-45% authorized immigrants (Kuka et al., 2020; Pope, 2016). Since DACA is only available to unauthorized immigrants, this group's outcomes would bias any estimates toward zero. To ameliorate this bias, previous literature has implemented restrictive sample criteria as a robustness check. In our paper, however, the restricted sample is the starting point and main analytical sample, bringing our DDD estimates closer (albeit still with noise) to a representation of DACA's intention-to-treat effect (Amuedo-Dorantes and Wang, 2025).

The second contributor of bias is the fact that DACA takeup rates are well below 100%. For example, in 2018, approximately 54% of the DACA-eligible population applied and was approved for DACA (Mira, 2022). If our sample perfectly screened for eligibility (it does not), we could feasibly estimate an average treatment effect on the treated by scaling our DDD coefficients by $1/\lambda$, where λ is the average takeup rate for DACA-eligible individuals over our sample period. Using the 2018 rate of 54% implies that the magnitudes of DACA's ATT effects could be almost double what our estimates suggest. Further, the likelihood of significant contamination of the treatment group with authorized immigrants, still present even in our restricted sample, suggests

an even greater scaling factor. In lieu of further back-of-the-envelope attempts in the direction of cleaner ITT and ATT estimates, we run our main DDD regressions on two subsamples with above average takeup rates. One contains those from countries-of-origin with DACA takeup above 40%, as of 2017.²³ The second subsample restricts to only Mexican-born individuals, since this group had the highest DACA takeup rate of all (65% in 2017). In Appendix tables A7, A8, and A9, the triple-difference coefficients are consistent in sign and larger in magnitude for these subsamples when compared to the main estimation. This suggests that given the assumptions of our model, the true ATT effects of DACA amid increasing enforcement action are likely higher than our main estimates suggest.

In a related exercise, we examine whether our results hold in commuting zones where enforcement and DACA are most economically relevant, splitting commuting zones by baseline Hispanic population share. In commuting zones above the 75th percentile of baseline Hispanic share, which contain the large majority of our sample, the triple-interaction estimates closely mirror the baseline: positive and statistically significant for income outcomes (Table A11), negative and statistically significant for below-poverty status and SNAP participation (Table A12), and positive but significant only at the 10 percent level for labor force participation (Table A10). Commuting zones below the 25th percentile contain too few individuals in our sample to yield informative estimates.

This spatial concentration is consistent with differences in treatment intensity and economic relevance across local labor markets. Because legal status is not directly observed, the empirical design captures ITT effects, which are naturally attenuated in areas where the share of likely-undocumented or DACA-eligible individuals is small. In high-Hispanic-share CZs, a larger fraction of the population is directly exposed to immigration enforcement and eligible for DACA, increasing the effective treatment intensity and the likelihood of detecting economically meaningful responses. Moreover, immigration enforcement is more salient in these areas, both because of greater visibility and stronger social network effects, amplifying behavioral responses to policy changes.

C. Robustness and Falsification

1. Alternative Specifications

We run several robustness checks regarding specification. First, we examine the sensitivity of our results to using state-by-year fixed effects instead of year fixed effects, which control for state-level time-varying shocks such as policy changes, economic conditions, and labor market dynamics that

²³Mexico, El Salvador, Honduras, Brazil, Peru, and Ecuador. Statistics from <https://www.migrationpolicy.org/programs/data-hub/deferred-action-childhood-arrivals-daca-profiles>, accessed for January 2018 via WaybackMachine.

may be correlated with both immigration enforcement and outcomes. Panel A of Tables [A13](#), [A14](#), and [A15](#) shows that the estimates remain very similar to the baseline across all outcome categories. For employment, the triple interaction during the DACA period remains around 0.013 to 0.015, while for income it remains positive and statistically significant, with magnitudes comparable to the baseline. For poverty outcomes, the triple interaction continues to show increases in the poverty ratio and reductions in poverty and SNAP participation, with coefficients similar in size and significance to the main results.

Second, we add CZ-specific linear time trends, which account for differential underlying trends in local labor markets that may evolve independently of immigration enforcement. Panel B of Tables [A13](#), [A14](#), and [A15](#) shows that the estimates remain stable across all outcomes. The employment results continue to show small positive effects during the DACA period, income effects remain positive and statistically significant, and poverty results continue to indicate improvements for DACA-eligible individuals in higher enforcement areas.

Third, we replace year-by-arrival-cohort fixed effects with year-by-age-at-immigration fixed effects, which flexibly control for cohort differences based on age at arrival rather than year of entry. Panel C of Tables [A13](#), [A14](#), and [A15](#) shows that the main results remain largely unchanged. Across all outcomes, the magnitudes and statistical significance of the triple interaction terms are consistent with the baseline, indicating that the findings are not sensitive to how cohort effects are specified.

2. Alternative Sample and Control Group Definitions

We next examine whether our results are sensitive to how we construct the analysis sample and define the control group, since both rely on choices, an eligibility window, a Hispanic restriction, a residency threshold, and a residual method for approximating undocumented status, that are not dictated by the data alone. We run four checks. The first restricts the sample to those with at least five years of U.S. residence, typically the maximum duration of a student visa, to exclude likely temporary legal residents ([Amuedo-Dorantes and Antman, 2017](#)). This check (Tables [A16-A18](#), Panel A) leaves the income results consistent with the baseline, with all four income measures remaining positive and statistically significant, while the employment and poverty estimates retain their signs but mostly lose significance given the smaller sample.

The remaining three checks redefine the control group by isolating each eligibility margin separately, restricting the comparison group in turn to those who fail only the arrival-age criterion, only the residence-length criterion, or only the age-in-2012 criterion, while holding the other two at their eligible values. Panel B is the most informative, as it addresses a compositional concern: individuals failing only the residence requirement arrived after 2007 and enter our age window gradually, shifting the baseline comparison group toward recent arrivals after 2012. Restricting to

those who arrived at ages 16 or 17 yields a comparison group with stable composition, and the income results hold and slightly strengthen, while the poverty estimates keep their signs with less precision. Panel C leaves too small a comparison group to be informative. Panel D yields positive employment effects but income estimates that reverse sign, likely reflecting age-profile differences inherent in comparing older ineligible individuals with the eligible group (Slusky, 2017).

Overall, our income results are robust to the restriction that most directly addresses compositional change in the comparison group, while the employment and poverty results are directionally consistent but less precise. The divergent estimates in Panel D represent a limitation, which we attribute partly to the age differences inherent in that comparison. We therefore view the income effects as our most robust findings and the poverty effects as more suggestive.

3. Falsification Tests

As a falsification exercise, we re-estimate the baseline specification using naturalized Hispanic citizens, a group not directly exposed to immigration enforcement in the same way and ineligible for DACA.²⁴ Tables A19-A21 show the results.

During the DACA period (2013-2017), triple interaction terms are small and statistically insignificant across labor market, poverty, SNAP, and income outcomes, consistent with no differential effect while DACA was securely in place. In the post-2017 period, however, we find negative and statistically significant triple interaction coefficients for labor force participation and for several income measures, including logged income and total income. The income result is particularly notable because its sign matches, rather than contradicts, the negative (though insignificant) coefficient we estimate for the same period and outcome in our main sample, raising the possibility that both groups responded to a common shock during 2018-2019 rather than one specific to DACA eligibility.

We interpret this post-2017 pattern with caution. Naturalized citizens are not a fully clean placebo group: naturalization is not randomly assigned, and naturalized Hispanic citizens in high-enforcement commuting zones plausibly remain exposed to enforcement through mixed-status households and social networks. Recent work shows DACA affected naturalization decisions and household linkages among noncitizens connected to DACA-eligible individuals (Amuedo-Dorantes and Wang, 2025; Battaglia and Pearson, 2026), suggesting these ties are not fixed over our sample period. The 2018-2019 period also coincided with a broader escalation in immigration enforcement salience nationally, including the reinstatement of Secure Communities, which may have generated community-level economic effects unrelated to DACA eligibility specifically.

²⁴We construct a pseudo-eligibility indicator, labeled Pseudo-DACA Eligible in Tables A19-A21, using the same statutory thresholds as our main sample: arrival before age 16, age under 31 as of June 2012, at least five years of US residence by 2012, and the same education restriction, applied to naturalized citizens in place of the non-citizen restriction used to define the main analysis sample.

Given the short window and smaller falsification sample, we also cannot rule out sampling variation given the number of outcomes examined.

Together, these results temper but do not overturn our main interpretation. The absence of any triple interaction pattern during the DACA period, when our main buffering effect is strongest, is inconsistent with the concern that our baseline estimates simply capture broader trends or confounding local conditions over that period. The post-2017 pattern instead suggests that some of the weakening we document in DACA's protective effect may reflect general deterioration in conditions facing Hispanic communities in high-enforcement areas, rather than the erosion of DACA's credibility alone, a limitation of the falsification test rather than of the main design.

D. Disentangling the Buffering Effects of DACA

Having shown that DACA eligibility attenuated the economic costs of enforcement, and that this attenuation weakened once the program's future became uncertain, we next examine how this buffering effect operated. DACA's grant of work authorization and deferred action mechanically reduces both deportation risk and barriers to formal employment for eligible individuals, and Section 1-3 already document the resulting labor market and income gains. Here we investigate the more specific mechanisms through which that protection translates into the patterns we observe, rather than whether legal status itself matters. We consider three candidate channels: occupational upgrading into less immigrant-intensive work, geographic sorting into lower-enforcement commuting zones, and a psychological channel operating through reduced deportation fear. We are able to directly test the first two channels with our data; the third, for which we lack psychological or health measures, is discussed as a plausible complement rather than a formally tested channel. Across the two channels we can test, the evidence is consistent with DACA-eligible individuals being protected in place rather than through relocation, though we treat the occupational upgrading result as suggestive given its statistical precision.

1. Occupational Upgrading

A natural channel through which DACA eligibility could attenuate the costs of enforcement is occupational sorting rather than, or in addition to, changes in overall employment. Work authorization allows recipients to seek jobs outside the informal, often immigrant-intensive segments of the labor market where undocumented workers are concentrated and where raids, audits, and employer-based screening such as E-verify are most likely to bind. If DACA-eligible individuals use their new authorization to move into more formal or less immigrant-intensive occupations, particularly where local enforcement risk is high, this reallocation would appear as a shift in occupational composition even absent large changes in the extensive margin of employment documented

in Table 2.

To examine this channel, we construct an indicator equal to one if an individual works in an immigrant-intensive occupation and estimate our baseline triple-differences specification, conditional on being employed.²⁵ Table A22 reports the results.

Two patterns emerge. First, DACA-eligible individuals are already 3.4 percentage points less likely to work in an immigrant-intensive occupation than their just-ineligible counterparts in the pre-DACA baseline, a difference that is statistically significant at the 1 percent level and reflects persistent compositional differences between the two groups that predate the program. Second, and more central to our interpretation, the triple interaction between enforcement, DACA eligibility, and the DACA period is negative and statistically significant at the 10 percent level: a one standard deviation increase in enforcement intensity is associated with a 1.9 percentage point larger decline in the probability that DACA-eligible individuals work in an immigrant-intensive occupation, relative to their ineligible counterparts, during the years the program was fully in force. This pattern is consistent with an occupational upgrading channel, in which eligible individuals draw on work authorization to exit immigrant-intensive employment precisely where enforcement makes such jobs riskiest.

As with our main results, this differential effect does not persist once DACA's future became uncertain. The corresponding triple interaction in the post-2017 period is small and statistically indistinguishable from zero (-0.006). This attenuation mirrors the pattern documented for income and poverty outcomes in Sections 2 and 3, and reinforces our broader finding that it is the credibility of legal protection, rather than eligibility alone, that underlies DACA's buffering effect. These results also complement recent evidence that DACA's labor market effects operate partly through compositional shifts into safer, better-paying, and licensed occupations (Chin et al., 2026; Villanueva Kiser and Wilson, 2024) rather than through uniform wage gains within job type; our findings suggest that this reallocation is itself sensitive to local enforcement intensity, and concentrated in the period when DACA's protections were most credible.

2. In-Place Protection or Geographic Escape?

The buffering effect we document may reflect DACA shielding recipients in place, with work authorization and deportation deferral directly reducing the economic costs of enforcement where eligible individuals already live and work. It could also arise from geographic sorting, if eligible individuals relocated to lower-enforcement areas following DACA approval, in which case the

²⁵We classify an occupation, based on 2010 Census/ACS occupation codes, as immigrant-intensive if its noncitizen employment share falls at or above the 75th percentile of the weighted distribution across occupations. Occupation-level noncitizen shares are computed using the full ACS sample of prime-age (25 to 54) employed workers in the pre-DACA period (2005 to 2011), weighted by ACS person weights, so that the classification is fixed prior to our sample period and does not mechanically respond to enforcement or DACA eligibility in later years.

estimated buffering effect would reflect relocation rather than direct protection. Distinguishing between these explanations matters, since a program that protects in place carries different welfare implications than one that operates through residential sorting.

We construct individuals' prior-year location using the ACS measure of residence one year ago and probabilistically map both current and prior locations to CZs using standard crosswalks.²⁶ We then examine four migration outcomes: whether the individual moved in the past year, whether the individual moved to a lower-enforcement CZ, the same outcome conditional on moving, and the enforcement intensity of the current CZ.

We first estimate these outcomes as a function of DACA eligibility and period alone. Table A23, Panel A, reports little evidence that DACA eligibility systematically affects migration behavior: column (1) shows no meaningful change in the probability of moving, columns (2) and (3) show no consistent evidence that DACA-eligible individuals relocate to lower-enforcement CZs, and column (4) shows no systematic difference in current enforcement exposure by eligibility. However, this average comparison is compatible with differential sorting occurring specifically among individuals originating in high-enforcement CZs, since that subgroup's effect would be diluted once pooled with individuals who had little reason to relocate.

To test this directly, we re-estimate the migration outcomes using our baseline triple-difference specification from Equation (3), interacting enforcement exposure with DACA eligibility and period. Table A23, Panel B, reports these estimates. Across columns (5) through (7), the triple interaction between enforcement, DACA eligibility, and period is small in magnitude and statistically insignificant in every specification, indicating that migration away from high-enforcement CZs does not occur at differentially higher rates among DACA-eligible individuals. Column (8) shows a small but marginally significant interaction, though this outcome reflects overall enforcement exposure rather than migration behavior directly and is therefore less informative about sorting.

Overall, these results make geographic sorting into lower-enforcement areas an unlikely explanation for our main findings, though they cannot rule out selective departure from the United States, which our data do not capture.

3. Psychological Channel

Our tested mechanisms leave room for an additional channel operating through deportation fear rather than the probability of removal itself. Existing evidence supports both sides of this channel.

²⁶Because the ACS reports location at the PUMA level, we link individuals' current and prior-year residences to CZs using established PUMA-to-CZ crosswalks. Individuals are fractionally allocated across CZs based on population shares. For prior-year locations, where direct crosswalk weights are unavailable, we assume equal probability across feasible mappings before applying the standard PUMA-to-CZ allocation. The resulting regression weights combine the survey sampling weights with all mapping shares used to probabilistically assign individuals from prior residence to current PUMAs and then to CZs, ensuring that observations remain population-representative after all transformations.

Enforcement generates fear and avoidance behavior even among individuals not directly at risk, reducing safety net participation (Alsan and Yang, 2024; Watson, 2014) and worsening mental health (Luo and Kostandini, 2023), while DACA eligibility improves recipients' mental health and sleep (Giuntella and Lonsky, 2020; Giuntella et al., 2021). If fear depresses earnings independent of removal risk, DACA could offset enforcement's costs partly by relieving it, and the strength of this relief would depend on the credibility of DACA's protections, consistent with the buffering effect weakening after 2017. Our data contain no psychological or health measures, so we cannot test this channel directly.

VI. Conclusion

In this study, we test whether DACA eligibility shielded individuals from the economic costs of enhanced immigration enforcement. Since previous work has highlighted the spillover effects of enforcement to non-targeted groups, it is plausible that DACA-eligible individuals armed with the program's protections may still have been vulnerable to negative economic impacts (Alsan and Yang, 2024). Using a triple-differences approach, we uncover the differential effects of enforcement between those eligible for DACA and a similar but just-ineligible control group.

We find that DACA eligibility offset much of enforcement's adverse effect on earnings and poverty. For a one standard deviation increase in enforcement intensity during the years DACA was fully in force, eligible individuals were 1.9 percentage points less likely to be in poverty and had 4.1 percent higher wage income relative to the just-ineligible comparison group, offsetting roughly 75 to 90 percent of the enforcement penalty on income and poverty. This protection weakened after the 2017 rescission attempt, consistent with the credibility of legal protection mattering for its value. To our knowledge, these are the first estimates of the interaction between DACA and immigration enforcement policy, and they are robust to alternative specifications and to most alternative sample and comparison-group constructions. While data limitations keep us from directly testing a psychological mechanism, we find evidence of occupational upgrading out of immigrant-intensive jobs as one channel, and show that our results are not driven by residential sorting into low-enforcement areas.

Our study has implications for other research pursuits related to US-based immigration enforcement. In the absence of controls for DACA eligibility or relevant sample restrictions, estimates of the impact of immigration enforcement policy on the undocumented may be biased toward zero due to the presence of the DACA-eligible, a group that we have shown has a distinct response to enforcement. Researchers should take care, to the extent the data allow, that their designs address this potential source of bias.

Finally, our paper suggests considerations for public policy. Our research design using the

interaction of a protective policy with an array of punitive ones is itself an artifact of the convoluted priorities of immigration law in the United States. As of this writing, federal funding and efforts have shifted dramatically toward enforcement (East et al., 2026). In response, some states and localities have passed protective measures for targeted residents (Chishti and Bush-Joseph, 2026). Regarding DACA itself, the second Trump administration has encouraged remaining recipients to self-deport, and has arrested and deported more than 170 current DACA holders (Olivares, 2026). As the program's intended federal protections may no longer hold, it is vital for policymakers concerned with the impacts of enforcement, especially those at subnational levels, to understand the substantial effects of in-place protection. With DACA's uncertain path through courts and public opinion still ongoing, our study provides a new accounting of its downstream impacts to communities living amid fear of deportation.

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VII. Figures and Tables

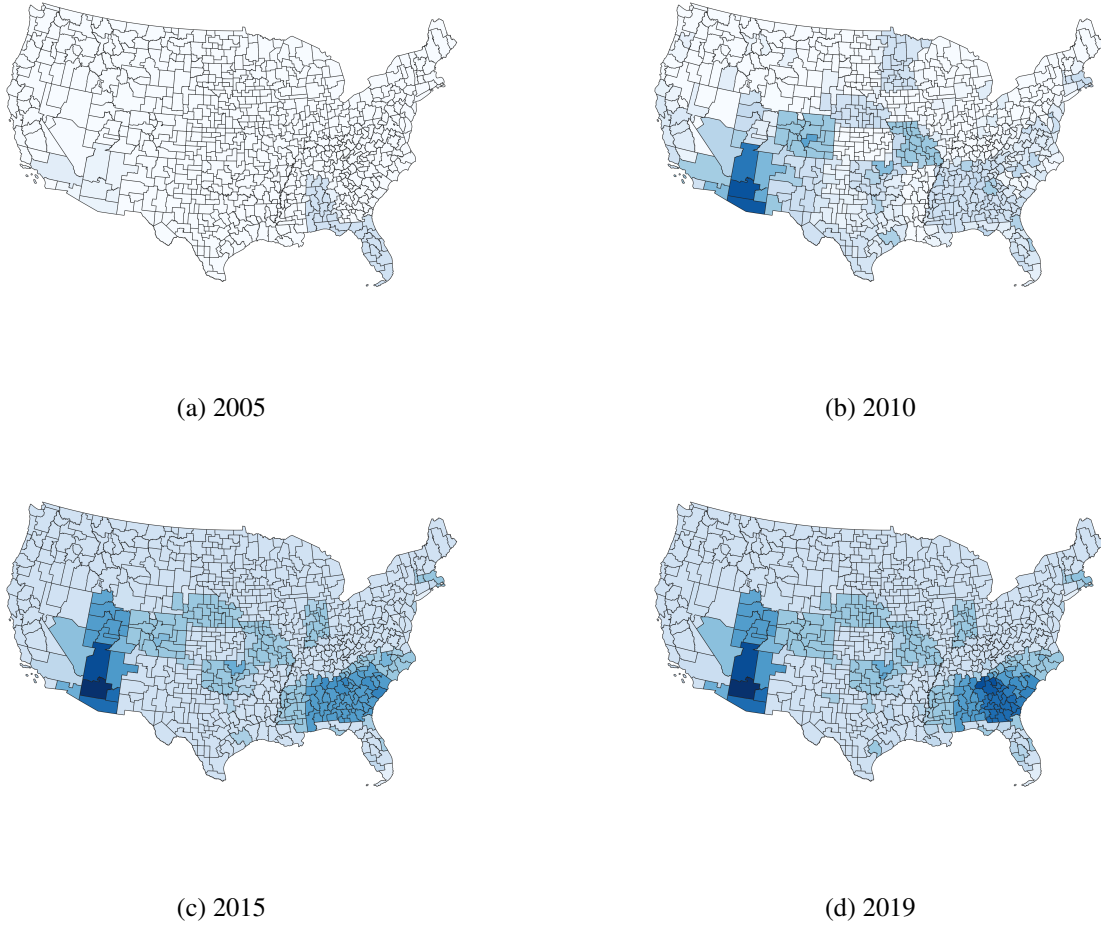
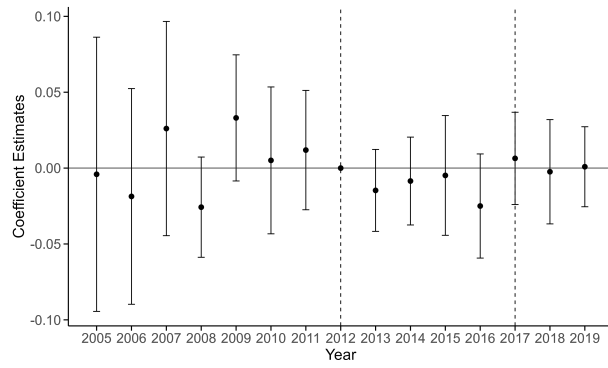
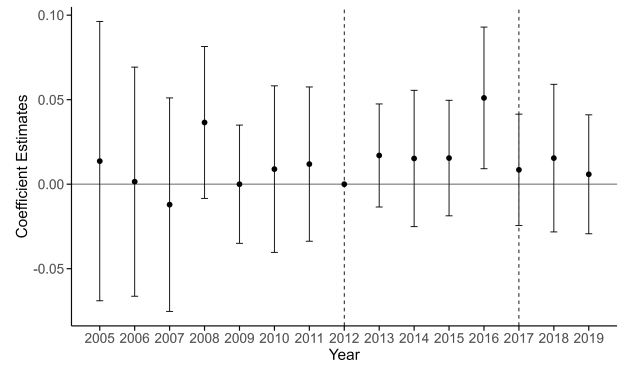


Figure 1: Evolution of Interior Immigration Enforcement Intensity.

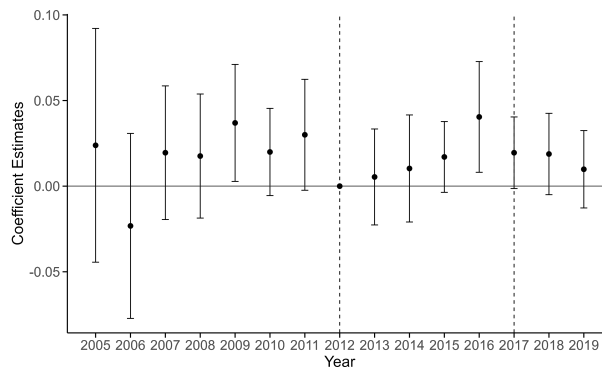
Notes: Maps generated by the authors. Shows color-graded enforcement intensity (light to dark) by commuting zone by year. Sources as described in Section III.



(a) Not in the Labor Force

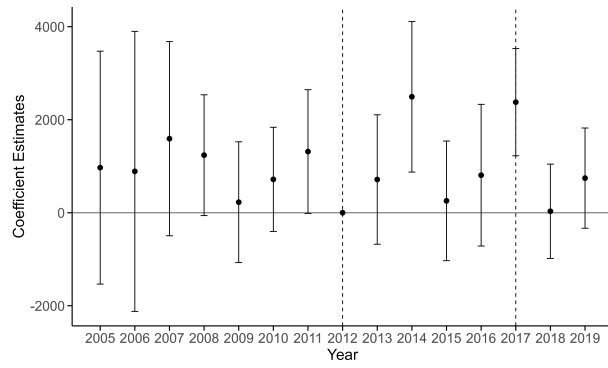


(b) Employed

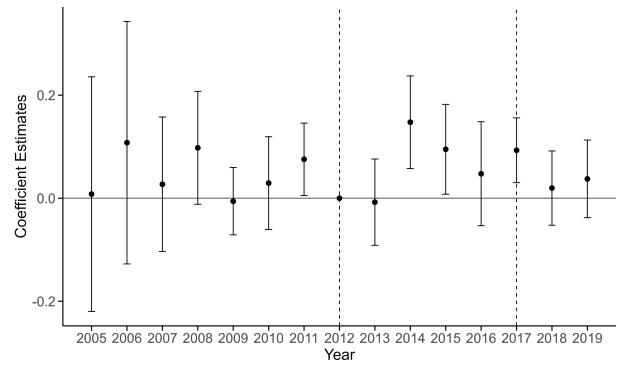


(c) Employed (Conditional on Being in the Labor Force)

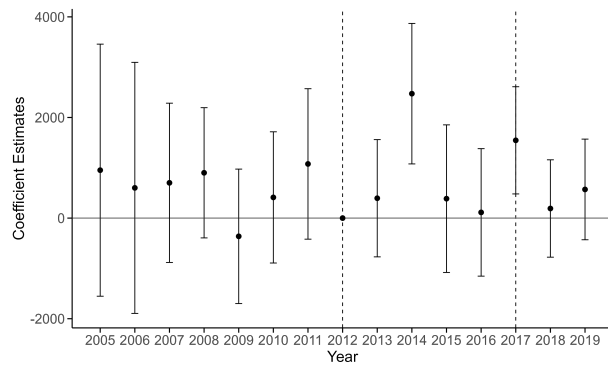
Figure 2: Event Study of Immigration Enforcement $\times$ DACA Eligibility on Employment Outcomes



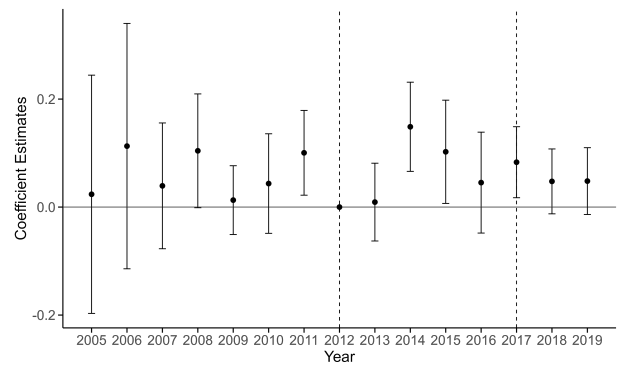
(a) Total Wage Income



(b) Total Logged Wage Income

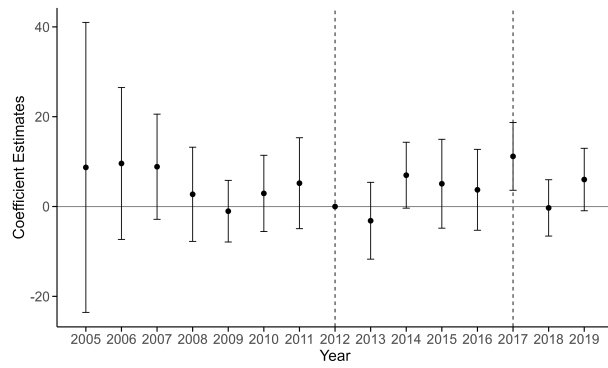


(c) Total Personal Income

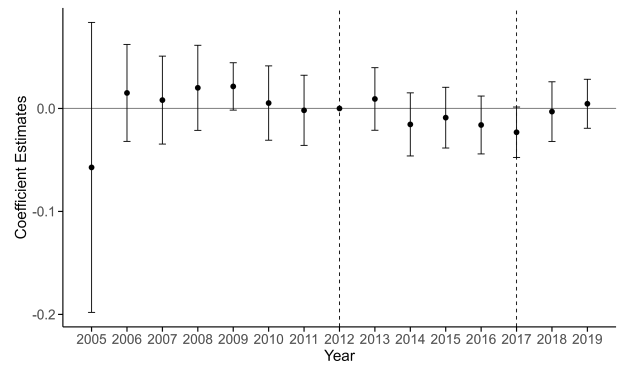


(d) Logged Total Personal Income

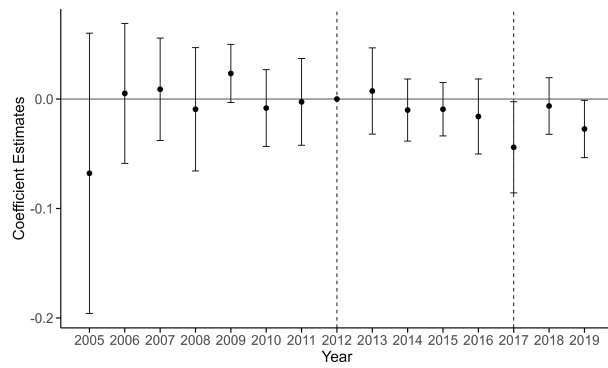
Figure 3: Event Study of Immigration Enforcement $\times$ DACA Eligibility on Income



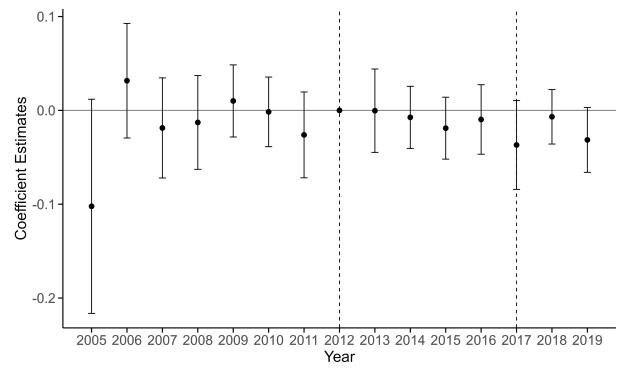
(a) Income-to-Poverty Ratio



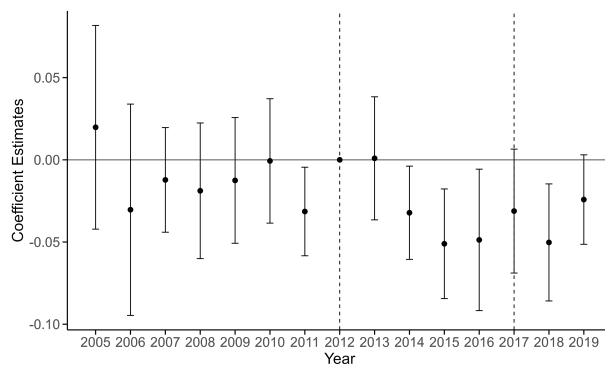
(b) Income below Poverty Line



(c) Income below 125% of Poverty Line



(d) Income below 150% of Poverty Line



(e) SNAP Participation

Figure 4: Event Study of Immigration Enforcement $\times$ DACA Eligibility on Poverty and Welfare Outcomes

Table 1: Summary Statistics by DACA Eligibility and Period

Variable	Pre-DACA Period				Post-DACA Period			
	DACA Ineligible		DACA Eligible		DACA Ineligible		DACA Eligible	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
In labor force	0.75	0.43	0.70	0.46	0.72	0.45	0.78	0.41
Working (all)	0.68	0.47	0.61	0.49	0.67	0.47	0.73	0.45
Working (conditional)	0.91	0.29	0.87	0.34	0.93	0.26	0.93	0.26
Wage income	15,232.07	15,239.99	13,120.39	15,278.52	16,148.91	17,525.46	18,292.54	18,046.76
Total personal income	16,102.87	15,382.06	13,838.19	15,547.39	17,368.07	17,806.71	19,368.47	18,249.54
Income-to-poverty ratio	192.37	123.53	197.14	124.53	204.82	131.81	219.51	132.95
Below poverty line	0.24	0.43	0.23	0.42	0.22	0.42	0.19	0.39
Below 125% poverty line	0.33	0.47	0.32	0.47	0.31	0.46	0.27	0.44
Below 150% poverty line	0.42	0.49	0.41	0.49	0.39	0.49	0.35	0.48
SNAP participation	0.17	0.37	0.16	0.37	0.21	0.41	0.20	0.40
Age	23.09	3.21	22.09	3.13	25.40	5.64	25.43	4.83
No schooling	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
Male	0.500	0.500	0.553	0.497	0.571	0.495	0.614	0.487
No high school	0.01	0.10	0.00	0.07	0.01	0.08	0.00	0.05
Some high school	0.05	0.23	0.08	0.28	0.08	0.28	0.04	0.20
High school	0.75	0.43	0.65	0.48	0.68	0.47	0.63	0.48
Single	0.67	0.47	0.76	0.43	0.63	0.48	0.66	0.47
Married	0.26	0.44	0.17	0.38	0.29	0.46	0.26	0.44
Absent spouse	0.06	0.23	0.05	0.22	0.06	0.23	0.06	0.23
Divorced/Widowed	0.01	0.12	0.01	0.12	0.02	0.15	0.03	0.16
Number of children	0.55	0.94	0.39	0.82	0.77	1.20	0.69	1.14
Years in US	9.72	3.06	16.65	5.43	5.44	5.19	14.06	5.32
Year of immigration	2002.29	3.06	1995.36	5.43	2006.57	5.19	1997.94	5.32
Age at immigration	16.53	0.53	8.41	4.68	15.72	1.75	7.38	4.56
Observations	14,345		56,542		21,405		77,662	
Unique individual-years	11,571		46,044		15,495		55,340	

Note: Means and standard deviations are reported. The pre-period covers 2005 to 2012, and the post-period covers 2013 to 2019. All statistics are weighted using survey weights that adjust for sampling and for the allocation of individuals to CZs.

Table 2: Effects of Immigration Enforcement and DACA Eligibility on Labor Market Participation

	In Labor Force	Employed (All)	Employed (Conditional)
Enforcement	-0.008 (0.008)	-0.006 (0.010)	0.002 (0.006)
DACA Eligible	-0.007 (0.006)	-0.010 (0.007)	-0.007 (0.007)
Enforcement $\times$ DACA Eligible	-0.002 (0.005)	-0.006 (0.008)	-0.008 (0.007)
Enforcement $\times$ DACA Active	-0.004 (0.012)	-0.006 (0.015)	-0.005 (0.008)
DACA Eligible $\times$ DACA Active	0.037*** (0.011)	0.048*** (0.011)	0.025*** (0.009)
Enforcement $\times$ DACA Uncertain	0.005 (0.016)	0.007 (0.016)	0.002 (0.007)
DACA Eligible $\times$ DACA Uncertain	0.014 (0.012)	0.011 (0.013)	-0.002 (0.007)
Enforcement $\times$ DACA Eligible $\times$ DACA Active	0.014* (0.008)	0.016 (0.012)	0.008 (0.009)
Enforcement $\times$ DACA Eligible $\times$ DACA Uncertain	0.006 (0.012)	0.005 (0.014)	0.003 (0.008)
CZ Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Year $\times$ Cohort Fixed Effects	Yes	Yes	Yes
<i>N</i>	169,954	169,954	120,875
Unique individual-years	128,450	128,450	93,068
Adjusted R^2	0.151	0.154	0.046

Notes: Each column reports estimates from a separate regression examining the interaction between immigration enforcement exposure and DACA eligibility. Column (1) examines labor force participation, while columns (2) and (3) examine employment outcomes for the full sample and conditional on labor force participation. All regressions include commuting zone fixed effects, year fixed effects, and year-by-immigration cohort fixed effects, along with individual demographic controls and local economic controls. Observations are weighted using ACS person weights and standard errors are clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Effects of Immigration Enforcement and DACA Eligibility on Income Outcomes (Conditional on Being Employed)

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
Enforcement	7.266 (202.977)	-0.002 (0.015)	-94.925 (206.803)	-0.009 (0.013)
DACA Eligible	843.808*** (247.703)	0.053*** (0.015)	844.396*** (263.100)	0.046*** (0.015)
Enforcement $\times$ DACA Eligible	-156.548 (199.579)	-0.001 (0.013)	-208.135 (196.602)	0.001 (0.011)
Enforcement $\times$ DACA Active	-709.665*** (228.036)	-0.041*** (0.014)	-498.940** (216.361)	-0.025* (0.015)
DACA Eligible $\times$ DACA Active	663.209* (388.885)	0.017 (0.022)	797.900** (394.681)	0.028 (0.021)
Enforcement $\times$ DACA Uncertain	5.199 (251.361)	-0.002 (0.019)	-159.801 (252.240)	0.002 (0.016)
DACA Eligible $\times$ DACA Uncertain	1192.358 (748.460)	0.005 (0.033)	1211.876 (784.947)	0.028 (0.036)
Enforcement $\times$ DACA Eligible $\times$ DACA Active	687.678** (266.356)	0.041*** (0.015)	667.456** (258.606)	0.029* (0.016)
Enforcement $\times$ DACA Eligible $\times$ DACA Uncertain	-225.090 (271.740)	-0.006 (0.025)	81.523 (300.143)	-0.001 (0.020)
CZ Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Year $\times$ Cohort Fixed Effects	Yes	Yes	Yes	Yes
<i>N</i>	109,637	105,096	109,637	109,621
Unique individual-years	83,925	80,401	83,925	83,910
Adjusted R^2	0.201	0.247	0.238	0.239

Notes: Each column reports estimates from a separate regression examining the interaction between immigration enforcement exposure and DACA eligibility. All regressions include commuting zone fixed effects, year fixed effects, and year-by-immigration cohort fixed effects, along with individual demographic controls and local economic controls. Observations are weighted using ACS person weights and standard errors are clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects of Immigration Enforcement and DACA Eligibility on Poverty and Welfare Outcomes

	Poverty Ratio	Below Poverty Line	Below 125% Poverty	Below 150% Poverty	SNAP Participation
Enforcement	-1.699 (1.447)	0.010 (0.007)	0.005 (0.008)	0.006 (0.009)	-0.016* (0.008)
DACA Eligible	6.759*** (1.923)	-0.024*** (0.008)	-0.014* (0.007)	-0.020** (0.009)	-0.014** (0.007)
Enforcement $\times$ DACA Eligible	-0.318 (1.245)	-0.004 (0.006)	0.005 (0.005)	0.004 (0.006)	0.009* (0.005)
Enforcement $\times$ DACA Active	-2.025 (1.721)	0.015* (0.008)	0.012 (0.010)	0.003 (0.010)	0.017** (0.008)
DACA Eligible $\times$ DACA Active	3.049 (2.271)	-0.001 (0.009)	-0.019* (0.010)	-0.019* (0.011)	-0.009 (0.010)
Enforcement $\times$ DACA Uncertain	0.626 (2.363)	-0.006 (0.008)	-0.002 (0.011)	-0.003 (0.012)	0.015 (0.010)
DACA Eligible $\times$ DACA Uncertain	5.950 (3.919)	0.028*** (0.011)	0.020 (0.013)	0.006 (0.017)	0.021 (0.018)
Enforcement $\times$ DACA Eligible $\times$ DACA Active	3.247* (1.806)	-0.019*** (0.006)	-0.020** (0.009)	-0.012 (0.012)	-0.017** (0.007)
Enforcement $\times$ DACA Eligible $\times$ DACA Uncertain	1.275 (2.199)	-0.006 (0.006)	-0.022** (0.009)	-0.016 (0.011)	-0.019** (0.010)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year $\times$ Cohort Fixed Effects	Yes	Yes	Yes	Yes	Yes
<i>N</i>	169,954	169,954	169,954	169,954	169,954
Unique individual-years	128,450	128,450	128,450	128,450	128,450
Adjusted R^2	0.168	0.085	0.102	0.116	0.099

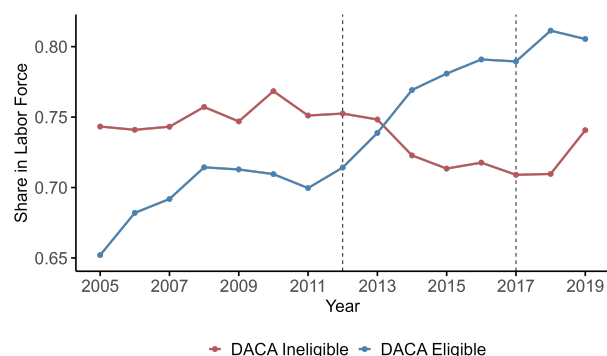
Notes: Each column reports estimates from a separate regression examining the interaction between immigration enforcement exposure and DACA eligibility. All regressions include commuting zone fixed effects, year fixed effects, and year-by-immigration cohort fixed effects, along with individual demographic controls and local economic controls. Observations are weighted using ACS person weights and standard errors are clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix

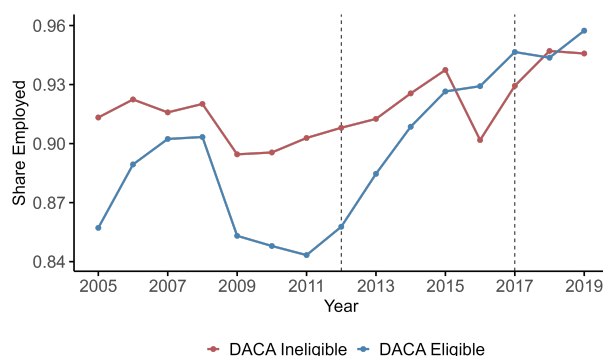
Temporary Legal Protection as a Buffer against Immigration Enforcement:
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Online Appendix: Additional Figures and Tables

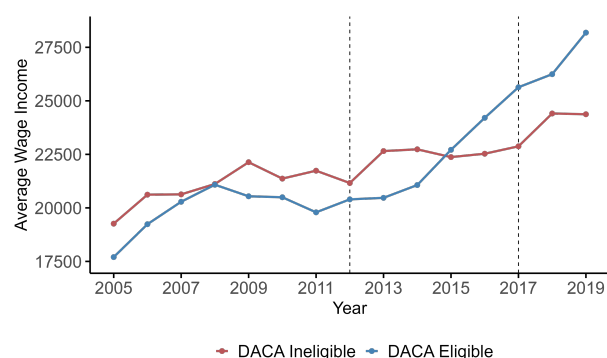


(a) Share in Labor Force

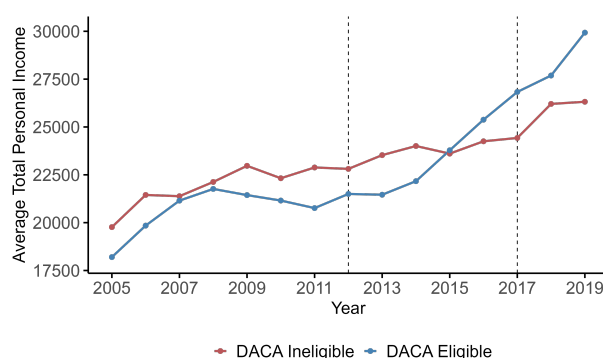


(b) Share Employed (Conditional on Being in Labor Force)

Figure A1: Trends between DACA Ineligible and DACA Eligible Individuals over Time on Employment Outcomes

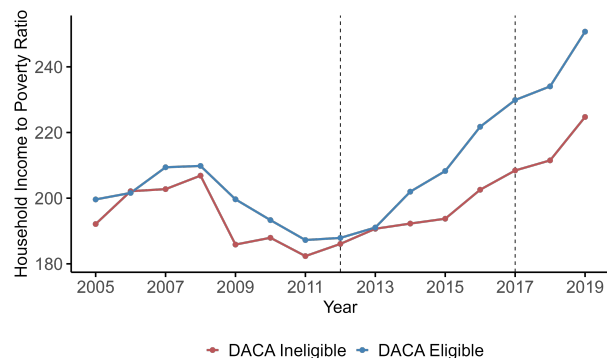


(a) Average Wage Income

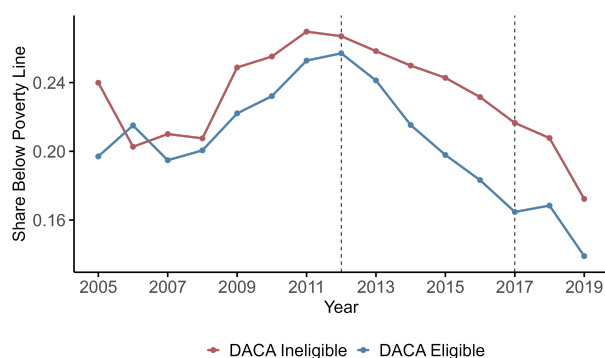


(b) Average Total Personal Income

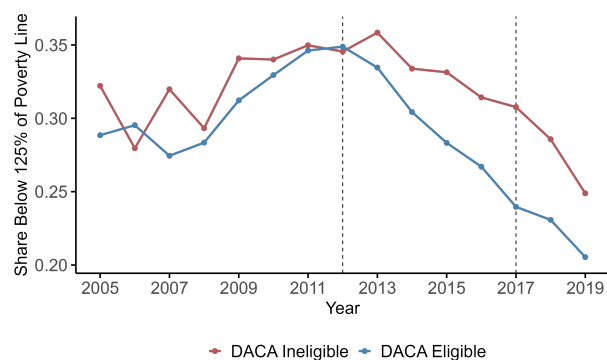
Figure A2: Trends between DACA Ineligible and DACA Eligible Individuals over Time on Income Outcomes



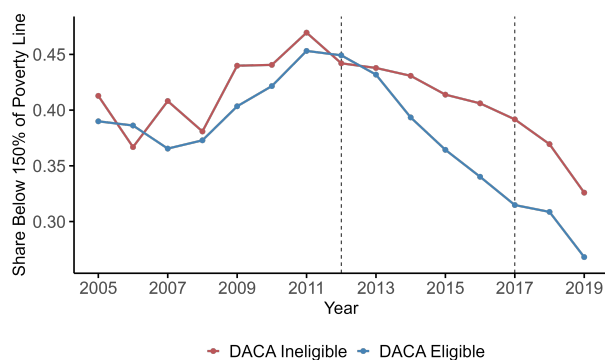
(a) Household Income to Poverty Ratio



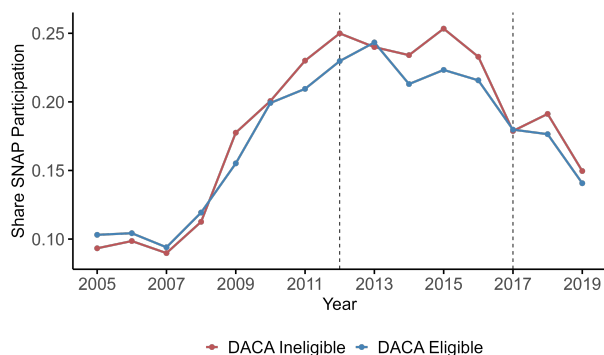
(b) Share Below Poverty Line



(c) Share Below 125% of Poverty Line



(d) Share Below 150% of Poverty Line



(e) Share SNAP Participation

Figure A3: Trends between DACA Ineligible and DACA Eligible Individuals over Time on Poverty and Welfare Outcomes

Table A1: Total Effect of Immigration Enforcement on Labor Market Participation by DACA Eligibility and Period

	In Labor Force	Employed (All)	Employed (Conditional)
<i>Panel A: Ineligible Group, DACA Active Years (2013–2017)</i>			
Enforcement	-0.012 (0.007)	-0.013 (0.008)	-0.003 (0.006)
<i>Panel B: Ineligible Group, DACA Uncertain Year (2018–2019)</i>			
Enforcement	-0.003 (0.011)	0.000 (0.010)	0.004 (0.005)
<i>Panel C: Eligible Group, DACA Active Years (2013–2017)</i>			
Enforcement	-0.000 (0.003)	-0.002 (0.003)	-0.003 (0.003)
<i>Panel D: Eligible Group, DACA Uncertain Year (2018–2019)</i>			
Enforcement	0.000 (0.004)	-0.000 (0.005)	-0.001 (0.004)
CZ Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Year $\times$ Cohort Fixed Effects	Yes	Yes	Yes
<i>N</i>	169,954	169,954	120,875
Unique individual-years	128,450	128,450	93,068
Adjusted R^2	0.151	0.154	0.046

Notes: Each panel reports the total effect of immigration enforcement on labor market outcomes for a given group and period, computed as a linear combination of coefficients from the baseline triple-difference specification in Table 2. Panel A (B) reports the enforcement effect for the ineligible group during (after) the DACA period. Panel C (D) reports the total enforcement effect for DACA-eligible individuals during (after) the DACA period. Standard errors are computed via the delta method and clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Effects of Immigration Enforcement and DACA Eligibility on Income Outcomes (Not Conditional on Being Employed)

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
Enforcement	-298.5 (214.5)	-0.00525 (0.0158)	-371.4* (196.6)	-0.0157 (0.0148)
DACA Eligible	554.5*** (202.9)	0.0554*** (0.0137)	548.5*** (207.0)	0.0554*** (0.0137)
Enforcement x DACA Eligible	55.09 (176.5)	0.00264 (0.0148)	4.998 (177.1)	0.0121 (0.0133)
Enforcement x During DACA	-361.0 (276.3)	-0.0447*** (0.0161)	-134.9 (306.9)	-0.0174 (0.0165)
DACA Eligible x During DACA	1099.4*** (331.3)	0.0236 (0.0217)	1283.5*** (337.2)	0.0269 (0.0222)
Enforcement x After DACA	307.1 (306.6)	0.00551 (0.0208)	131.6 (342.7)	0.0231 (0.0165)
DACA Eligible x After DACA	1284.9** (562.8)	-0.0104 (0.0311)	1324.1** (601.9)	-0.000215 (0.0317)
Enforcement x DACA Eligible x During DACA	413.1 (293.0)	0.0355** (0.0159)	314.4 (296.4)	0.00903 (0.0190)
Enforcement x DACA Eligible x After DACA	-372.8 (302.3)	-0.0186 (0.0271)	-90.63 (373.1)	-0.0299 (0.0200)
CZ Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Year × Cohort Fixed Effects	Yes	Yes	Yes	Yes
<i>N</i>	169,950	119,978	169,950	127,985
Adjusted R-squared	0.242	0.247	0.268	0.233

Notes: Each column reports estimates from a separate regression examining the interaction between immigration enforcement exposure and DACA eligibility. All regressions include commuting zone fixed effects, year fixed effects, and year-by-immigration cohort fixed effects, along with individual demographic controls and local economic controls. Observations are weighted using ACS person weights and standard errors are clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Differences in DDD Coefficients Across Active and Uncertain Periods

Outcome	Difference	Std. Error	95% CI	p-value
<i>Panel A: Employment</i>				
Labor Force Participation	0.008	(0.009)	[-0.009, 0.025]	0.351
Employed (All)	0.011	(0.010)	[-0.008, 0.030]	0.255
Employed (Conditional)	0.005	(0.007)	[-0.010, 0.020]	0.489
<i>Panel B: Income</i>				
Wage Income	912.768***	(275.938)	[371.015, 1454.522]	0.001
Log Wage Income	0.046**	(0.023)	[0.002, 0.091]	0.041
Total Personal Income	585.933**	(278.156)	[39.825, 1132.040]	0.036
Log Total Personal Income	0.030	(0.022)	[-0.013, 0.074]	0.174
<i>Panel C: Poverty</i>				
Poverty Ratio	1.972	(2.300)	[-2.544, 6.488]	0.392
Below Poverty Line	-0.014**	(0.006)	[-0.026, -0.001]	0.031
Below 125% FPL	0.002	(0.009)	[-0.015, 0.019]	0.854
Below 150% FPL	0.005	(0.010)	[-0.016, 0.025]	0.660
SNAP Participation	0.002	(0.009)	[-0.016, 0.020]	0.831

Notes: Panels A, B, and C correspond to the exact regressions in Tables 2, 3, and 4, respectively. The differences reported here are the linear combinations of coefficients $\beta_8 - \beta_9$, as in Equation 3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Total Effect of Immigration Enforcement on Income Outcomes by DACA Eligibility and Period

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
<i>Panel A: Ineligible Group, DACA Active Years (2013–2017)</i>				
Enforcement	-702.398*** (207.354)	-0.043*** (0.013)	-593.865*** (219.104)	-0.034*** (0.013)
<i>Panel B: Ineligible Group, DACA Uncertain Year (2018–2019)</i>				
Enforcement	12.466 (204.698)	-0.004 (0.014)	-254.726 (197.538)	-0.007 (0.013)
<i>Panel C: Eligible Group, DACA Active Years (2013–2017)</i>				
Enforcement	-171.268 (167.453)	-0.004 (0.010)	-134.544 (166.594)	-0.004 (0.009)
<i>Panel D: Eligible Group, DACA Uncertain Year (2018–2019)</i>				
Enforcement	-369.172** (165.892)	-0.012 (0.010)	-381.337** (167.844)	-0.007 (0.010)
CZ Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Year × Cohort Fixed Effects	Yes	Yes	Yes	Yes
<i>N</i>	109,637	105,096	109,637	109,621
Unique individual-years	83,925	80,401	83,925	83,910
Adjusted <i>R</i> ²	0.201	0.247	0.238	0.239

Notes: Each panel reports the total effect of immigration enforcement on income outcomes for a given group and period, computed as a linear combination of coefficients from the baseline triple-difference specification in Table 3. Panel A (B) reports the enforcement effect for the ineligible group during (after) the DACA period. Panel C (D) reports the total enforcement effect for DACA-eligible individuals during (after) the DACA period. All regressions condition on being employed. Standard errors are computed via the delta method and clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Total Effect of Immigration Enforcement on Poverty and Welfare Outcomes by DACA Eligibility and Period

	Poverty Ratio	Below Poverty Line	Below 125% Poverty	Below 150% Poverty	SNAP
<i>Panel A: Ineligible Group, DACA Active Years (2013–2017)</i>					
Enforcement	-3.724*** (1.417)	0.026*** (0.005)	0.017** (0.007)	0.009 (0.007)	0.001 (0.006)
<i>Panel B: Ineligible Group, DACA Uncertain Year (2018–2019)</i>					
Enforcement	-1.073 (2.045)	0.004 (0.006)	0.003 (0.008)	0.003 (0.009)	-0.001 (0.007)
<i>Panel C: Eligible Group, DACA Active Years (2013–2017)</i>					
Enforcement	-0.794 (1.283)	0.002 (0.004)	0.001 (0.005)	0.002 (0.007)	-0.007 (0.005)
<i>Panel D: Eligible Group, DACA Uncertain Year (2018–2019)</i>					
Enforcement	-0.116 (1.437)	-0.006 (0.004)	-0.014** (0.006)	-0.009 (0.008)	-0.011** (0.005)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year $\times$ Cohort Fixed Effects	Yes	Yes	Yes	Yes	Yes
<i>N</i>	169,954	169,954	169,954	169,954	169,954
Unique individual-years	128,450	128,450	128,450	128,450	128,450
Adjusted R^2	0.168	0.085	0.102	0.116	0.099

Notes: Each panel reports the total effect of immigration enforcement on poverty and welfare outcomes for a given group and period, computed as a linear combination of coefficients from the baseline triple-difference specification in Table 4. Panel A (B) reports the enforcement effect for the ineligible group during (after) the DACA period. Panel C (D) reports the total enforcement effect for DACA-eligible individuals during (after) the DACA period. Standard errors are computed via the delta method and clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Implied Share of the Enforcement Effect Offset by DACA Eligibility, DACA-Active Period (2013-2017)

Outcome	Implied Offset	95% CI	
In Labor Force	98.2%	[46.3%, 150.2%]	†
Employed (All)	82.3%	[39.6%, 125.0%]	†
Employed (Conditional)	-3.6%	[-304.9%, 297.6%]	‡
Wage Income	75.6%	[32.3%, 118.9%]	
Log Wage Income	90.3%	[46.6%, 133.9%]	
Total Personal Income	77.3%	[28.8%, 125.9%]	
Log Total Personal Income	87.4%	[36.1%, 138.8%]	
Poverty Ratio	78.7%	[13.6%, 143.7%]	†
Below Poverty Line	91.3%	[65.1%, 117.6%]	
Below 125% Poverty	91.8%	[30.2%, 153.4%]	†
Below 150% Poverty	82.5%	[-65.0%, 230.0%]	‡
SNAP Participation	—	—	‡

Notes: Each row reports the implied percentage of the total enforcement effect offset by DACA eligibility during the active DACA period, computed as one minus the ratio of the total enforcement effect for DACA-eligible individuals to the total enforcement effect for ineligible individuals, using the coefficients from Tables 2-4. Confidence intervals are computed via the delta method. † indicates a wide confidence interval that should be interpreted with caution. ‡ indicates a confidence interval that includes zero or is otherwise uninformative; these estimates should not be interpreted as evidence of any offset. Post-2017 (DACA-uncertain period) estimates are omitted from this table because the underlying enforcement effect on the ineligible group becomes imprecisely estimated in that period for all outcomes, making the implied offset ratio uninformative throughout.

Table A7: High-Takeup Sample Estimates, Employment

	In Labor Force		Employed (All)		Employed (Conditional)	
	(1) High-Takeup	(2) Mexico	(3) High-Takeup	(4) Mexico	(5) High-Takeup	(6) Mexico
Enforcement	-0.00498 (0.00819)	-0.00667 (0.00833)	-0.000790 (0.0100)	-0.00222 (0.0106)	0.00504 (0.00671)	0.00460 (0.00756)
DACA Eligible	-0.00339 (0.00659)	-0.000974 (0.00759)	-0.00483 (0.00765)	0.00159 (0.00803)	-0.00362 (0.00785)	0.00369 (0.00881)
Enforcement x DACA Eligible	-0.00551 (0.00463)	-0.00648 (0.00467)	-0.0132* (0.00755)	-0.0134* (0.00794)	-0.0135* (0.00749)	-0.0127 (0.00799)
Enforcement × DACA Active	-0.00828 (0.0121)	-0.00996 (0.0139)	-0.0115 (0.0150)	-0.00964 (0.0164)	-0.00627 (0.00806)	-0.000965 (0.00720)
DACA Eligible × DACA Active	0.0296*** (0.0106)	0.0178 (0.0116)	0.0393*** (0.0118)	0.0280** (0.0126)	0.0218** (0.00991)	0.0196* (0.0113)
Enforcement × DACA Uncertain	-0.00112 (0.0158)	0.00129 (0.0152)	-0.00299 (0.0156)	-0.00161 (0.0155)	-0.00321 (0.00538)	-0.00387 (0.00641)
DACA Eligible × DACA Uncertain	0.00874 (0.0124)	0.00297 (0.0132)	0.00660 (0.0144)	-0.000668 (0.0157)	-0.00214 (0.00834)	-0.00602 (0.00923)
Enforcement x DACA Eligible × DACA Active	0.0200** (0.00807)	0.0239** (0.0105)	0.0243* (0.0130)	0.0240 (0.0148)	0.0113 (0.00965)	0.00566 (0.00865)
Enforcement x DACA Eligible × DACA Uncertain	0.0121 (0.0115)	0.0135 (0.0109)	0.0174 (0.0136)	0.0195 (0.0132)	0.0112 (0.00700)	0.0123 (0.00833)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year x Cohort Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Unique Individual-Years	147,102	129,151	147,102	129,151	104,939	92,113
Adjusted R-squared	0.156	0.158	0.159	0.161	0.049	0.051

Notes: Specification mirrors the main estimation. The High-Takeup sample consists of individuals born in countries with estimated takeup over 40% in 2017 (see footnote in main text). The Mexico subsample consists of only individuals born in Mexico.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A8: High-Takeup Sample Estimates, Income (Conditional on Employed)

	Wage Income		Log Wage Income		Total Personal Income		Log Total Personal Income	
	(1) High-Takeup	(2) Mexico	(3) High-Takeup	(4) Mexico	(5) High-Takeup	(6) Mexico	(7) High-Takeup	(8) Mexico
Enforcement	23.98 (229.2)	202.3 (255.7)	-0.00103 (0.0168)	0.00486 (0.0174)	-67.32 (233.2)	24.85 (258.0)	-0.00420 (0.0153)	0.00143 (0.0156)
DACA Eligible	810.0*** (262.0)	629.3** (287.4)	0.0479*** (0.0163)	0.0458** (0.0180)	885.6*** (270.8)	757.1** (293.5)	0.0373** (0.0170)	0.0374* (0.0197)
Enforcement x DACA Eligible	-92.34 (217.4)	-130.5 (246.2)	0.00182 (0.0157)	0.00814 (0.0165)	-201.2 (219.8)	-141.6 (250.6)	-0.000580 (0.0134)	0.00543 (0.0147)
Enforcement × DACA Active	-837.0*** (310.0)	-1219.1*** (320.3)	-0.0453** (0.0184)	-0.0673*** (0.0173)	-578.6** (257.5)	-819.2*** (294.1)	-0.0334* (0.0177)	-0.0558*** (0.0163)
DACA Eligible × DACA Active	782.0* (429.7)	751.4 (540.8)	0.0261 (0.0258)	0.0178 (0.0289)	873.6** (436.7)	697.2 (518.8)	0.0369 (0.0254)	0.0239 (0.0284)
Enforcement × DACA Uncertain	-7.123 (341.5)	-279.2 (378.9)	-0.00183 (0.0228)	-0.00397 (0.0252)	-237.7 (336.2)	-355.2 (368.4)	0.00132 (0.0197)	-0.00399 (0.0216)
DACA Eligible × DACA Uncertain	1424.4* (838.1)	1049.0 (1010.3)	0.0188 (0.0352)	0.0120 (0.0427)	1203.8 (868.7)	795.9 (999.4)	0.0402 (0.0402)	0.0185 (0.0474)
Enforcement x DACA Eligible × DACA Active	867.3*** (280.2)	1170.5*** (308.5)	0.0476*** (0.0178)	0.0631*** (0.0187)	824.8*** (267.9)	1021.6*** (324.8)	0.0400** (0.0171)	0.0549*** (0.0172)
Enforcement x DACA Eligible × DACA Uncertain	-129.8 (304.5)	51.86 (358.4)	-0.00880 (0.0258)	-0.0153 (0.0280)	235.1 (317.9)	282.5 (362.4)	-0.00274 (0.0201)	-0.00585 (0.0225)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year x Cohort Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Unique Individual-Years	95,261	83,645	91,316	80,173	95,261	83,645	95,251	83,636
Adjusted R-squared	0.202	0.201	0.247	0.244	0.240	0.240	0.239	0.237

Notes: Specification mirrors the main estimation. The High-Takeup sample consists of individuals born in countries with estimated takeup over 40% in 2017 (see footnote in main text). The Mexico subsample consists of only individuals born in Mexico.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A9: High-Takeup Sample Estimates, Poverty

	Poverty Ratio		Below Poverty Line		Below 150% Poverty		SNAP Participation	
	(1) High-Takeup	(2) Mexico	(3) High-Takeup	(4) Mexico	(5) High-Takeup	(6) Mexico	(7) High-Takeup	(8) Mexico
Enforcement	-0.792 (1.689)	-1.284 (1.942)	0.00384 (0.00797)	0.00613 (0.00903)	0.00286 (0.00972)	0.00777 (0.0106)	-0.0169* (0.00930)	-0.0162 (0.0104)
DACA Eligible	5.564*** (2.007)	5.124** (2.150)	-0.0206** (0.00858)	-0.0205** (0.00995)	-0.0129 (0.0100)	-0.0131 (0.0110)	-0.0137* (0.00749)	-0.0150* (0.00810)
Enforcement x DACA Eligible	-0.811 (1.344)	0.0285 (1.477)	-0.0000543 (0.00546)	-0.00196 (0.00539)	0.00760 (0.00766)	0.00363 (0.00810)	0.0110** (0.00548)	0.00809 (0.00554)
Enforcement × DACA Active	-2.835* (1.621)	-3.616* (1.913)	0.0207** (0.00949)	0.0192* (0.0101)	0.00784 (0.0113)	0.00723 (0.0126)	0.0186* (0.00951)	0.0174* (0.00968)
DACA Eligible × DACA Active	5.165** (2.513)	3.812 (2.711)	-0.00759 (0.0110)	-0.00578 (0.0127)	-0.0265** (0.0121)	-0.0212 (0.0134)	-0.0132 (0.0108)	-0.0195* (0.0115)
Enforcement × DACA Uncertain	-0.442 (2.241)	0.267 (2.466)	-0.00297 (0.00675)	-0.00512 (0.00712)	0.00251 (0.0119)	0.00241 (0.0136)	0.00994 (0.00957)	0.0108 (0.0108)
DACA Eligible × DACA Uncertain	4.374 (3.993)	5.563 (4.447)	0.0306** (0.0125)	0.0269* (0.0142)	0.00521 (0.0197)	0.00892 (0.0225)	0.0145 (0.0179)	0.0109 (0.0194)
Enforcement x DACA Eligible × DACA Active	3.523** (1.707)	4.622** (1.879)	-0.0232*** (0.00719)	-0.0214*** (0.00722)	-0.0140 (0.0131)	-0.0133 (0.0144)	-0.0205** (0.00875)	-0.0185** (0.00893)
Enforcement x DACA Eligible × DACA Uncertain	2.394 (1.974)	0.976 (2.073)	-0.00841 (0.00679)	-0.00454 (0.00735)	-0.0201* (0.0114)	-0.0163 (0.0122)	-0.0164* (0.00842)	-0.0139 (0.00892)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year x Cohort Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Unique Individual-Years	147,102	129,151	147,102	129,151	147,102	129,151	147,102	129,151
Adjusted R-squared	0.173	0.177	0.088	0.091	0.119	0.123	0.102	0.106

Notes: Specification mirrors the main estimation. The High-Takeup sample consists of individuals born in countries with estimated take-up over 40% in 2017 (see footnote in main text). The Mexico subsample consists of only individuals born in Mexico.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Heterogeneity by Baseline Hispanic Share: Employment Outcomes

	In Labor Force	Employed (All)	Employed (Conditional)
<i>Panel A: CZs below the 25th percentile of baseline Hispanic share</i>			
Enforcement	0.009 (0.079)	0.039 (0.078)	-0.002 (0.060)
DACA Eligible	0.040 (0.095)	-0.038 (0.084)	-0.096 (0.059)
Enforcement $\times$ DACA Eligible	0.018 (0.086)	0.034 (0.076)	0.047 (0.050)
Enforcement $\times$ DACA Active	-0.030 (0.137)	-0.076 (0.123)	-0.020 (0.096)
DACA Eligible $\times$ DACA Active	0.107 (0.124)	0.150 (0.113)	0.065 (0.058)
Enforcement $\times$ DACA Uncertain	-0.071 (0.179)	-0.119 (0.172)	-0.038 (0.084)
DACA Eligible $\times$ DACA Uncertain	0.124 (0.132)	0.229* (0.118)	0.152** (0.076)
Enforcement $\times$ DACA Eligible $\times$ DACA Active	0.008 (0.142)	-0.005 (0.133)	-0.082 (0.097)
Enforcement $\times$ DACA Eligible $\times$ DACA Uncertain	0.118 (0.194)	0.132 (0.185)	-0.030 (0.089)
<i>N</i>	3281	3281	2088
Unique individual-years	1963	1963	1253
Adjusted R^2	0.359	0.385	0.472
<i>Panel B: CZs above the 75th percentile of baseline Hispanic share</i>			
Enforcement	-0.009 (0.009)	-0.008 (0.011)	-0.000 (0.007)
DACA Eligible	-0.008 (0.007)	-0.008 (0.008)	-0.002 (0.008)
Enforcement $\times$ DACA Eligible	-0.003 (0.006)	-0.006 (0.009)	-0.007 (0.008)
Enforcement $\times$ DACA Active	-0.004 (0.013)	-0.005 (0.016)	-0.004 (0.009)
DACA Eligible $\times$ DACA Active	0.038*** (0.012)	0.049*** (0.013)	0.024** (0.010)
Enforcement $\times$ DACA Uncertain	0.001 (0.018)	0.004 (0.018)	0.002 (0.007)
DACA Eligible $\times$ DACA Uncertain	0.020 (0.013)	0.014 (0.015)	-0.006 (0.007)
Enforcement $\times$ DACA Eligible $\times$ DACA Active	0.015* (0.009)	0.017 (0.014)	0.007 (0.010)
Enforcement $\times$ DACA Eligible $\times$ DACA Uncertain	0.014 (0.014)	0.013 (0.017)	0.004 (0.008)
<i>N</i>	132454	132454	94747
Unique individual-years	110585	110585	80240
Adjusted R^2	0.154	0.156	0.047

Notes: Each panel reports estimates from separate regressions split by baseline Hispanic share at the commuting zone level. All specifications include the same controls and fixed effects as the baseline model. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Heterogeneity by Baseline Hispanic Share: Income Outcomes

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
<i>Panel A: CZs below the 25th percentile of baseline Hispanic share</i>				
Enforcement	-3603.062 (3680.583)	-0.411* (0.218)	-4697.393 (3415.112)	-0.442** (0.221)
DACA Eligible	-4048.832 (3890.898)	0.057 (0.185)	-2361.126 (3235.434)	0.051 (0.176)
Enforcement × DACA Eligible	-628.735 (3827.113)	0.247 (0.218)	-193.809 (3450.136)	0.167 (0.219)
Enforcement × DACA Active	-6045.858 (5321.022)	-0.017 (0.464)	-2162.199 (5287.919)	0.111 (0.417)
DACA Eligible × DACA Active	4150.790 (5041.049)	-0.019 (0.229)	2349.073 (4519.557)	-0.062 (0.224)
Enforcement × DACA Uncertain	7291.611 (6565.803)	0.346 (0.318)	5185.148 (6217.129)	0.471 (0.323)
DACA Eligible × DACA Uncertain	8979.652 (6852.652)	-0.083 (0.283)	10499.096 (6452.657)	-0.013 (0.271)
Enforcement × DACA Eligible × DACA Active	4479.931 (6234.843)	-0.420 (0.541)	299.735 (6132.220)	-0.361 (0.450)
Enforcement × DACA Eligible × DACA Uncertain	-15312.380* (8509.181)	-0.760* (0.422)	-11951.134 (8333.854)	-0.641 (0.394)
<i>N</i>	1973	1903	1973	1973
Unique individual-years	1174	1128	1174	1174
Adjusted R^2	0.507	0.614	0.532	0.588
<i>Panel B: CZs above the 75th percentile of baseline Hispanic share</i>				
Enforcement	4.447 (212.039)	-0.002 (0.016)	-76.092 (219.149)	-0.011 (0.015)
DACA Eligible	1164.766*** (249.212)	0.073*** (0.015)	1137.138*** (270.411)	0.067*** (0.015)
Enforcement × DACA Eligible	-114.641 (200.916)	0.000 (0.014)	-203.325 (204.407)	0.002 (0.013)
Enforcement × DACA Active	-759.608*** (207.349)	-0.042*** (0.015)	-578.649** (225.675)	-0.025 (0.016)
DACA Eligible × DACA Active	58.604 (408.590)	-0.021 (0.025)	182.844 (405.646)	-0.011 (0.023)
Enforcement × DACA Uncertain	-65.277 (245.693)	-0.000 (0.021)	-271.106 (252.134)	0.003 (0.017)
DACA Eligible × DACA Uncertain	874.880 (796.118)	-0.006 (0.035)	1044.641 (862.270)	0.018 (0.039)
Enforcement × DACA Eligible × DACA Active	672.283** (265.128)	0.039** (0.016)	720.818*** (266.568)	0.029* (0.017)
Enforcement × DACA Eligible × DACA Uncertain	-231.097 (281.103)	-0.011 (0.026)	126.437 (329.159)	-0.003 (0.021)
<i>N</i>	85654	81974	85654	85639
Unique individual-years	72223	69107	72223	72209
Adjusted R^2	0.198	0.243	0.236	0.236

Notes: Each panel reports estimates from separate regressions split by baseline Hispanic share at the commuting zone level. All specifications include the same controls and fixed effects as the baseline model. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Heterogeneity by Baseline Hispanic Share: Poverty and SNAP Outcomes

	Poverty Ratio	Below Poverty Line	Below 125p Poverty Line	Below 150p Poverty Line	SNAP
<i>Panel A: CZs below the 25th percentile of baseline Hispanic share</i>					
Enforcement	6.324 (19.832)	0.021 (0.088)	0.122 (0.096)	-0.024 (0.089)	-0.030 (0.049)
DACA Eligible	6.977 (18.006)	0.047 (0.090)	0.063 (0.090)	-0.006 (0.075)	0.082* (0.048)
Enforcement × DACA Eligible	15.330 (23.914)	-0.018 (0.085)	-0.050 (0.091)	0.019 (0.081)	0.054 (0.059)
Enforcement × DACA Active	2.106 (33.897)	-0.032 (0.165)	-0.003 (0.172)	-0.111 (0.168)	0.022 (0.091)
DACA Eligible × DACA Active	-4.828 (26.794)	-0.127 (0.104)	-0.119 (0.114)	-0.054 (0.095)	-0.066 (0.067)
Enforcement × DACA Uncertain	-14.254 (52.517)	0.022 (0.148)	0.130 (0.201)	0.144 (0.184)	0.061 (0.123)
DACA Eligible × DACA Uncertain	75.815* (45.558)	-0.348** (0.148)	-0.251* (0.148)	-0.284** (0.131)	-0.086 (0.119)
Enforcement × DACA Eligible × DACA Active	8.115 (39.084)	-0.003 (0.168)	-0.025 (0.176)	0.049 (0.189)	0.071 (0.073)
Enforcement × DACA Eligible × DACA Uncertain	-11.262 (62.815)	-0.108 (0.182)	-0.178 (0.219)	-0.156 (0.221)	-0.009 (0.125)
<i>N</i>	3281	3281	3281	3281	3281
Unique individual-years	1963	1963	1963	1963	1963
Adjusted R^2	0.451	0.432	0.414	0.434	0.357
<i>Panel B: CZs above the 75th percentile of baseline Hispanic share</i>					
Enforcement	-2.333* (1.397)	0.014** (0.007)	0.008 (0.008)	0.009 (0.009)	-0.012 (0.009)
DACA Eligible	7.392*** (2.064)	-0.025*** (0.008)	-0.014* (0.008)	-0.023** (0.010)	-0.017** (0.007)
Enforcement × DACA Eligible	0.218 (1.216)	-0.008 (0.006)	0.003 (0.005)	0.002 (0.006)	0.007 (0.005)
Enforcement × DACA Active	-0.861 (1.578)	0.006 (0.007)	0.007 (0.011)	0.000 (0.010)	0.016* (0.009)
DACA Eligible × DACA Active	3.358 (2.441)	-0.001 (0.010)	-0.022* (0.012)	-0.023* (0.013)	-0.002 (0.011)
Enforcement × DACA Uncertain	-0.104 (2.822)	-0.009 (0.008)	-0.003 (0.013)	0.001 (0.014)	0.010 (0.010)
DACA Eligible × DACA Uncertain	8.040* (4.185)	0.028*** (0.011)	0.018 (0.014)	0.006 (0.020)	0.020 (0.020)
Enforcement × DACA Eligible × DACA Active	2.114 (1.743)	-0.013** (0.006)	-0.017* (0.010)	-0.010 (0.012)	-0.017** (0.008)
Enforcement × DACA Eligible × DACA Uncertain	1.945 (2.773)	-0.005 (0.006)	-0.021* (0.011)	-0.020 (0.014)	-0.016 (0.010)
<i>N</i>	132454	132454	132454	132454	132454
Unique individual-years	110585	110585	110585	110585	110585
Adjusted R^2	0.169	0.082	0.100	0.115	0.099

Notes: Each panel reports estimates from separate regressions split by baseline Hispanic share at the commuting zone level. All specifications include the same controls and fixed effects as the baseline model. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Robustness to Fixed Effects and Trends: Employment Outcomes

	In Labor Force	Employed (All)	Employed (Conditional)
<i>Panel A: State-by-Year Fixed Effects</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	0.013 (0.008)	0.015 (0.012)	0.007 (0.009)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	0.001 (0.012)	0.001 (0.015)	0.002 (0.008)
<i>N</i>	169,954	169,954	120,875
Unique individual-years	128,450	128,450	93,068
Adjusted R^2	0.155	0.159	0.053
<i>Panel B: CZ-Specific Linear Time Trends</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	0.011 (0.008)	0.014 (0.012)	0.008 (0.008)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	0.003 (0.012)	0.002 (0.015)	0.001 (0.009)
<i>N</i>	169,954	169,954	120,875
Unique individual-years	128,450	128,450	93,068
Adjusted R^2	0.151	0.154	0.030
<i>Panel C: Year-by-Age-at-Immigration Fixed Effects</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	0.016* (0.009)	0.014 (0.013)	0.000 (0.008)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	0.006 (0.012)	0.001 (0.014)	-0.006 (0.008)
<i>N</i>	169,954	169,954	120,875
Unique individual-years	128,450	128,450	93,068
Adjusted R^2	0.150	0.153	0.046

Notes: Each panel reports estimates from separate regressions that modify or replace components of the fixed effects structure. All specifications include the same controls, commuting zone fixed effects, year fixed effects, and year-by-immigration-cohort fixed effects as in the baseline specification unless otherwise noted. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Robustness to Fixed Effects and Trends: Income Outcomes

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
<i>Panel A: State-by-Year Fixed Effects</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	743.727*** (260.758)	0.044*** (0.015)	682.768** (275.387)	0.032** (0.015)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-89.276 (288.820)	0.001 (0.027)	230.673 (339.195)	0.005 (0.021)
<i>N</i>	109,637	105,096	109,637	109,621
Unique individual-years	83,925	80,401	83,925	83,910
Adjusted R^2	0.205	0.252	0.242	0.244
<i>Panel B: CZ-Specific Linear Time Trends</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	715.113*** (260.730)	0.042*** (0.015)	721.446*** (266.685)	0.032** (0.016)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-156.647 (275.325)	-0.003 (0.025)	158.571 (307.271)	0.002 (0.019)
<i>N</i>	109,637	105,096	109,637	109,621
Unique individual-years	83,925	80,401	83,925	83,910
Adjusted R^2	0.200	0.247	0.237	0.239
<i>Panel C: Year-by-Age-at-Immigration Fixed Effects</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	671.846** (273.109)	0.038** (0.016)	600.819** (266.645)	0.027* (0.015)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-352.679 (265.582)	-0.014 (0.025)	-73.719 (258.418)	-0.006 (0.017)
<i>N</i>	109,637	105,096	109,637	109,621
Unique individual-years	83,925	80,401	83,925	83,910
Adjusted R^2	0.198	0.245	0.235	0.236

Notes: Each panel reports estimates from separate regressions that modify or replace components of the fixed effects structure. All specifications include the same controls, commuting zone fixed effects, year fixed effects, and year-by-immigration-cohort fixed effects as in the baseline specification unless otherwise noted. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A15: Robustness to Fixed Effects and Trends: Poverty Outcomes

	Poverty Ratio	Below Poverty Line	Below 125% Poverty Line	Below 150% Poverty Line	SNAP Participation
<i>Panel A: State-by-Year Fixed Effects</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	4.118** (1.955)	-0.020*** (0.006)	-0.020** (0.010)	-0.015 (0.013)	-0.016** (0.008)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	1.667 (2.446)	-0.004 (0.006)	-0.022** (0.010)	-0.019 (0.013)	-0.019* (0.010)
<i>N</i>	169,954	169,954	169,954	169,954	169,954
Unique individual-years	128,450	128,450	128,450	128,450	128,450
Adjusted R^2	0.174	0.092	0.108	0.123	0.105
<i>Panel B: CZ-Specific Linear Time Trends</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	3.929** (1.855)	-0.020*** (0.006)	-0.021** (0.010)	-0.015 (0.012)	-0.016** (0.007)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	1.989 (2.229)	-0.005 (0.006)	-0.023** (0.009)	-0.020* (0.012)	-0.017* (0.010)
<i>N</i>	169,954	169,954	169,954	169,954	169,954
Unique individual-years	128,450	128,450	128,450	128,450	128,450
Adjusted R^2	0.170	0.087	0.103	0.117	0.100
<i>Panel C: Year-by-Age-at-Immigration Fixed Effects</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	2.279 (1.742)	-0.016*** (0.006)	-0.015* (0.009)	-0.009 (0.011)	-0.022*** (0.008)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-0.052 (1.913)	-0.003 (0.006)	-0.017** (0.008)	-0.012 (0.010)	-0.022** (0.010)
<i>N</i>	169,954	169,954	169,954	169,954	169,954
Unique individual-years	128,450	128,450	128,450	128,450	128,450
Adjusted R^2	0.166	0.083	0.100	0.113	0.097

Notes: Each panel reports estimates from separate regressions that modify or replace components of the fixed effects structure. All specifications include the same controls, commuting zone fixed effects, year fixed effects, and year-by-immigration-cohort fixed effects as in the baseline specification unless otherwise noted. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Robustness to Sample Construction: Employment Outcomes

	In Labor Force	Employed (All)	Employed (Conditional)
<i>Panel A: ≥ 5 Years in U.S.</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	0.019 (0.016)	0.016 (0.019)	0.001 (0.007)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	0.002 (0.013)	0.009 (0.016)	0.011 (0.007)
<i>N</i>	154,643	154,643	111,597
Unique individual-years	117,269	117,269	86,183
Adjusted R^2	0.142	0.147	0.046
<i>Panel B: Redefining the Control Group: Isolating Ineligibility by Arrival-Age Criterion Only</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	0.023 (0.014)	0.020 (0.018)	0.001 (0.008)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	0.006 (0.011)	0.012 (0.014)	0.011 (0.007)
<i>N</i>	158,233	158,233	114,018
Unique individual-years	119,884	119,884	87,966
Adjusted R^2	0.143	0.148	0.046
<i>Panel C: Redefining the Control Group: Isolating Ineligibility by Residence Length Criterion Only</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	-0.061 (0.061)	-0.013 (0.046)	0.121 (0.092)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-0.041 (0.058)	-0.027 (0.047)	0.066 (0.090)
<i>N</i>	139,521	139,521	99,274
Unique individual-years	105,407	105,407	76,215
Adjusted R^2	0.143	0.149	0.051
<i>Panel D: Redefining the Control Group: Isolating Ineligibility by Age-in-2012 Criterion Only</i>			
Index $\times$ DACA Eligible $\times$ DACA Active	0.042*** (0.015)	0.043** (0.018)	0.010 (0.009)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-0.002 (0.019)	0.001 (0.016)	0.008 (0.007)
<i>N</i>	148,259	148,259	106,919
Unique individual-years	112,690	112,690	82,814
Adjusted R^2	0.136	0.141	0.048

Notes: Each panel reports estimates from separate regressions that vary the sample definition. All specifications include the same controls, commuting zone fixed effects, year fixed effects, and year-by-immigration-cohort fixed effects as in the baseline specification. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Robustness to Sample Construction: Income Outcomes

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
<i>Panel A: ≥ 5 Years in U.S.</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	857.905*** (310.600)	0.044*** (0.017)	683.833** (335.108)	0.031* (0.017)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	198.387 (389.680)	-0.008 (0.022)	608.096 (479.505)	-0.004 (0.023)
<i>N</i>	101,268	97,033	101,268	101,253
Unique individual-years	77,776	74,494	77,776	77,762
Adjusted R^2	0.196	0.243	0.234	0.235
<i>Panel B: Redefining the Control Group: Isolating Ineligibility by Arrival-Age Criterion Only</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	831.267*** (301.994)	0.050*** (0.017)	729.728** (302.964)	0.036** (0.018)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	155.224 (360.776)	-0.007 (0.021)	541.359 (373.515)	-0.000 (0.019)
<i>N</i>	103,472	99,137	103,472	103,456
Unique individual-years	79,383	76,021	79,383	79,368
Adjusted R^2	0.197	0.242	0.235	0.235
<i>Panel C: Redefining the Control Group: Isolating Ineligibility by Residence Length Criterion Only</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	745.987 (1720.292)	0.193 (0.279)	778.680 (1743.008)	0.187 (0.276)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	993.283 (1542.886)	0.266 (0.253)	1162.028 (1606.582)	0.273 (0.249)
<i>N</i>	89,581	86,077	89,581	89,569
Unique individual-years	68,379	65,704	68,379	68,368
Adjusted R^2	0.211	0.253	0.248	0.246
<i>Panel D: Redefining the Control Group: Isolating Ineligibility by Age-in-2012 Criterion Only</i>				
Index $\times$ DACA Eligible $\times$ DACA Active	-889.097 (543.425)	-0.010 (0.033)	-644.165 (453.718)	-0.005 (0.024)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-1789.122** (767.643)	-0.046* (0.028)	-1493.754** (731.135)	-0.009 (0.024)
<i>N</i>	96,890	92,802	96,890	96,874
Unique individual-years	74,660	71,518	74,660	74,645
Adjusted R^2	0.205	0.253	0.242	0.247

Notes: Each panel reports estimates from separate regressions that vary the sample definition. All specifications include the same controls and fixed effects as in the baseline model. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Robustness to Sample Construction: Poverty and Welfare Outcomes

	Poverty Ratio	Below Poverty Line	Below 125%	Below 150%	SNAP
<i>Panel A: ≥ 5 Years in U.S.</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	2.226 (3.051)	-0.010 (0.010)	-0.013 (0.016)	-0.010 (0.018)	-0.017* (0.010)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	3.120 (3.580)	0.002 (0.009)	-0.022** (0.009)	-0.019 (0.018)	-0.013* (0.008)
<i>N</i>	154,643	154,643	154,643	154,643	154,643
Unique individual-years	117,269	117,269	117,269	117,269	117,269
Adjusted R^2	0.175	0.086	0.105	0.120	0.101
<i>Panel B: Redefining the Control Group: Isolating Ineligibility by Arrival-Age Criterion Only</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	2.317 (2.811)	-0.013 (0.010)	-0.015 (0.016)	-0.012 (0.017)	-0.016* (0.008)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	1.829 (2.976)	0.003 (0.008)	-0.017* (0.009)	-0.014 (0.017)	-0.012 (0.008)
<i>N</i>	158,233	158,233	158,233	158,233	158,233
Unique individual-years	119,884	119,884	119,884	119,884	119,884
Adjusted R^2	0.174	0.085	0.104	0.119	0.101
<i>Panel C: Redefining the Control Group: Isolating Ineligibility by Residence Length Criterion Only</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	4.760 (13.302)	0.025 (0.059)	0.007 (0.058)	-0.006 (0.056)	-0.070** (0.033)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	6.564 (13.282)	0.042 (0.057)	-0.008 (0.058)	-0.032 (0.051)	-0.100*** (0.036)
<i>N</i>	139,521	139,521	139,521	139,521	139,521
Unique individual-years	105,407	105,407	105,407	105,407	105,407
Adjusted R^2	0.171	0.084	0.101	0.116	0.095
<i>Panel D: Redefining the Control Group: Isolating Ineligibility by Age-in-2012 Criterion Only</i>					
Index $\times$ DACA Eligible $\times$ DACA Active	-0.903 (2.337)	-0.002 (0.013)	-0.003 (0.011)	-0.004 (0.012)	0.005 (0.015)
Index $\times$ DACA Eligible $\times$ DACA Uncertain	-9.826** (4.878)	0.017* (0.010)	0.010 (0.010)	-0.002 (0.017)	0.009 (0.012)
<i>N</i>	148,259	148,259	148,259	148,259	148,259
Unique individual-years	112,690	112,690	112,690	112,690	112,690
Adjusted R^2	0.177	0.085	0.102	0.118	0.096

Notes: Each panel reports estimates from separate regressions that vary the sample definition. All specifications include the same controls, commuting zone fixed effects, year fixed effects, and year-by-immigration-cohort fixed effects as in the baseline specification. Standard errors are clustered at the commuting zone level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Falsification Test: Effects on Labor Market Outcomes (Naturalized Hispanic Citizens)

	In Labor Force	Employed (All)	Employed (Conditional)
Enforcement	-0.008 (0.012)	-0.009 (0.014)	-0.001 (0.008)
Pseudo-DACA Eligible	0.003 (0.019)	-0.006 (0.020)	-0.017 (0.011)
Enforcement $\times$ Pseudo-DACA Eligible	0.011 (0.013)	0.013 (0.014)	0.003 (0.007)
Enforcement $\times$ DACA Active	0.014 (0.015)	0.019 (0.018)	0.008 (0.009)
Pseudo-DACA Eligible $\times$ DACA Active	0.025 (0.020)	0.030 (0.021)	0.016 (0.012)
Enforcement $\times$ DACA Uncertain	0.027* (0.015)	0.020 (0.019)	-0.007 (0.011)
Pseudo-DACA Eligible $\times$ DACA Uncertain	-0.008 (0.024)	0.002 (0.027)	0.017 (0.014)
Enforcement $\times$ Pseudo-DACA Eligible $\times$ DACA Active	-0.007 (0.014)	-0.015 (0.016)	-0.012 (0.009)
Enforcement $\times$ Pseudo-DACA Eligible $\times$ DACA Uncertain	-0.029** (0.013)	-0.024 (0.018)	0.004 (0.012)
<i>N</i>	84,778	84,778	67,198
Unique individual-years	67,327	67,327	53,648
Adjusted R^2	0.101	0.105	0.039

Notes: Falsification test using naturalized Hispanic citizens. Specification mirrors the main model.

Table A20: Falsification Test: Effects on Income Outcomes (Naturalized Hispanic Citizens)

	Wage Income	Log Wage Income	Total Personal Income	Log Total Personal Income
Enforcement	-521.075 (610.924)	-0.041 (0.031)	-648.354 (568.891)	-0.035 (0.032)
Pseudo-DACA Eligible	990.601 (739.023)	0.008 (0.034)	738.175 (690.781)	0.016 (0.033)
Enforcement $\times$ Pseudo-DACA Eligible	650.063 (562.004)	0.035 (0.027)	578.394 (518.224)	0.029 (0.028)
Enforcement $\times$ DACA Active	265.556 (749.414)	0.046* (0.028)	597.882 (725.097)	0.032 (0.027)
Pseudo-DACA Eligible $\times$ DACA Active	1036.074 (1052.817)	0.056 (0.045)	1441.629 (987.349)	0.045 (0.042)
Enforcement $\times$ DACA Uncertain	151.076 (1034.967)	0.051 (0.038)	532.868 (924.160)	0.053 (0.038)
Pseudo-DACA Eligible $\times$ DACA Uncertain	2307.767* (1235.807)	0.056 (0.044)	2186.015** (993.285)	0.054 (0.045)
Enforcement $\times$ Pseudo-DACA Eligible $\times$ DACA Active	-723.251 (900.381)	-0.041 (0.035)	-884.836 (802.408)	-0.030 (0.034)
Enforcement $\times$ Pseudo-DACA Eligible $\times$ DACA Uncertain	-1374.305 (1124.815)	-0.078** (0.036)	-1658.906* (950.264)	-0.079** (0.036)
<i>N</i>	62,262	60,526	62,262	62,239
Unique individual-years	49,605	48,300	49,605	49,592
Adjusted R^2	0.246	0.295	0.270	0.294

Notes: Falsification test using naturalized Hispanic citizens. Specification mirrors the main model.

Table A21: Falsification Test: Poverty and Welfare Outcomes (Naturalized Hispanic Citizens)

	Poverty Ratio	Below Poverty Line	Below 125%	Below 150%	SNAP
Enforcement	2.725 (4.236)	-0.001 (0.009)	0.004 (0.009)	0.001 (0.013)	-0.014 (0.010)
Pseudo-DACA Eligible	-2.386 (5.017)	0.009 (0.012)	-0.007 (0.015)	-0.006 (0.015)	0.027** (0.013)
Enforcement × Pseudo-DACA Eligible	-0.178 (3.893)	0.003 (0.010)	0.003 (0.009)	0.001 (0.014)	0.007 (0.007)
Enforcement × DACA Active	5.078 (3.254)	-0.003 (0.011)	-0.010 (0.010)	-0.010 (0.013)	-0.010 (0.013)
Pseudo-DACA Eligible × DACA Active	4.918 (5.371)	-0.018 (0.014)	-0.002 (0.015)	0.004 (0.015)	-0.014 (0.017)
Enforcement × DACA Uncertain	-2.470 (5.434)	0.009 (0.015)	0.009 (0.013)	0.008 (0.016)	0.001 (0.012)
Pseudo-DACA Eligible × DACA Uncertain	2.477 (7.358)	-0.004 (0.016)	0.001 (0.019)	-0.001 (0.022)	-0.026 (0.019)
Enforcement × Pseudo-DACA Eligible × DACA Active	-2.158 (3.115)	-0.009 (0.010)	-0.006 (0.009)	-0.005 (0.013)	0.003 (0.013)
Enforcement × Pseudo-DACA Eligible × DACA Uncertain	3.059 (5.533)	-0.023 (0.016)	-0.021 (0.014)	-0.018 (0.020)	-0.004 (0.014)
<i>N</i>	84,778	84,778	84,778	84,778	84,778
Unique individual-years	67,327	67,327	67,327	67,327	67,327
Adjusted R^2	0.200	0.084	0.098	0.115	0.095

Notes: Falsification test using naturalized Hispanic citizens. Specification mirrors the main model.

Table A22: Effects of Immigration Enforcement and DACA Eligibility on Immigrant-Intensive Occupation (Conditional on Being Employed)

	Immigrant-Intensive Occupation
Enforcement	0.001 (0.008)
DACA Eligible	-0.034*** (0.010)
Enforcement $\times$ DACA Eligible	0.003 (0.008)
Enforcement $\times$ DACA Active	0.010 (0.010)
DACA Eligible $\times$ DACA Active	-0.017 (0.016)
Enforcement $\times$ DACA Uncertain	-0.011 (0.011)
DACA Eligible $\times$ DACA Uncertain	-0.040** (0.016)
Enforcement $\times$ DACA Eligible $\times$ DACA Active	-0.019* (0.011)
Enforcement $\times$ DACA Eligible $\times$ DACA Uncertain	-0.006 (0.013)
CZ Fixed Effects	Yes
Year Fixed Effects	Yes
Year $\times$ Cohort Fixed Effects	Yes
<i>N</i>	109,637
Unique individual-years	83,925
Adjusted R^2	0.137

Notes: The dependent variable is an indicator equal to one if the individual works in an immigrant-intensive occupation. The regression includes commuting zone fixed effects, year fixed effects, and year-by-immigration cohort fixed effects, along with individual demographic controls and local economic controls. Observations are weighted using ACS person weights and standard errors are clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Differential Migration Responses to Immigration Enforcement and DACA Eligibility

	Panel A: Baseline Specification				Panel B: With Enforcement Interaction			
	Moved Last Year	Moved to Lower Enf	Moved to Lower Enf (Conditional)	Current CZ Enf	Moved Last Year	Moved to Lower Enf	Moved to Lower Enf (Conditional)	Current CZ Enf
Enforcement					-0.002 (0.006)	0.003*** (0.001)	0.012*** (0.004)	0.888*** (0.001)
DACA Eligible	-0.009 (0.007)	-0.000 (0.000)	0.002* (0.001)	0.008 (0.009)	-0.008 (0.008)	-0.000 (0.000)	0.002 (0.001)	0.000 (0.000)
Enforcement × DACA Eligible					0.001 (0.005)	-0.000 (0.000)	0.000 (0.001)	0.000* (0.000)
Enforcement × DACA Active					-0.005 (0.007)	0.000 (0.000)	0.002 (0.002)	0.000 (0.000)
DACA Eligible × DACA Active	0.006 (0.011)	-0.000 (0.000)	-0.002 (0.001)	-0.017 (0.012)	0.005 (0.012)	0.000 (0.000)	-0.002 (0.002)	-0.000 (0.000)
Enforcement × DACA Uncertain					-0.004 (0.007)	-0.001 (0.001)	-0.002 (0.002)	0.001* (0.000)
DACA Eligible × DACA Uncertain	0.006 (0.018)	0.000 (0.000)	-0.000 (0.002)	-0.015 (0.015)	0.005 (0.019)	0.000 (0.000)	-0.001 (0.002)	-0.000 (0.000)
Enforcement × DACA Eligible × DACA Active					-0.001 (0.007)	0.000 (0.000)	-0.000 (0.001)	-0.000 (0.000)
Enforcement × DACA Eligible × DACA Uncertain					-0.002 (0.007)	0.000 (0.000)	0.002 (0.001)	-0.001* (0.000)
CZ Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year × Cohort Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	865,859	865,859	731,007	865,859	865,859	865,859	731,007	865,859
Unique individual-years	127,432	127,432	24,865	127,432	127,432	127,432	24,865	127,432
Adjusted <i>R</i> ²	0.036	0.046	0.186	0.847	0.036	0.047	0.191	0.999

Notes: Each column reports estimates from a separate regression examining migration responses and sorting behavior in relation to immigration enforcement and DACA eligibility. Columns (1) and (5) examine the probability of moving in the past year. Columns (2) and (6) examine whether individuals move to commuting zones with lower enforcement intensity. Columns (3) and (7) restrict this outcome to movers only. Columns (4) and (8) examine the enforcement intensity of the current commuting zone. Panel A reproduces the baseline specification; Panel B adds the standardized enforcement index and its interactions with DACA eligibility and period. All regressions include commuting zone fixed effects, year fixed effects, and year-by-immigration cohort fixed effects, along with individual demographic controls and local economic controls. Observations are weighted using survey weights combined with crosswalk-based allocation shares, and standard errors are clustered at the commuting zone level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.